MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

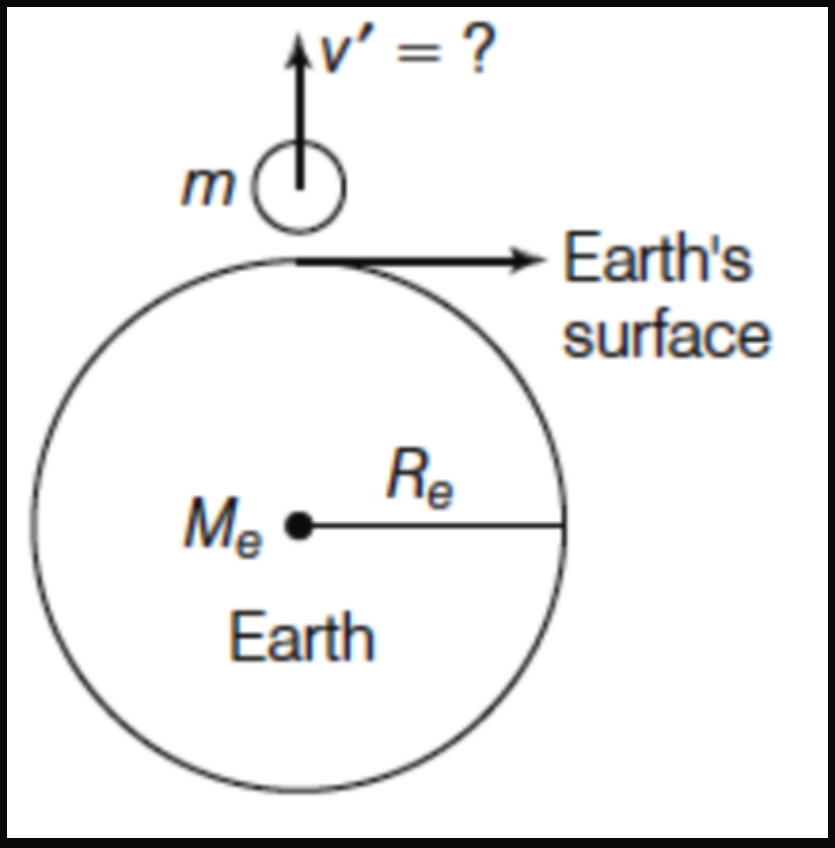
GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

What should be the speed by which we should throw a ball, so that it could reach upto a height $R_e/2$? (Ve = escape velocity from earth's surface)



(A) $\frac{v_e}{\sqrt{3}}$

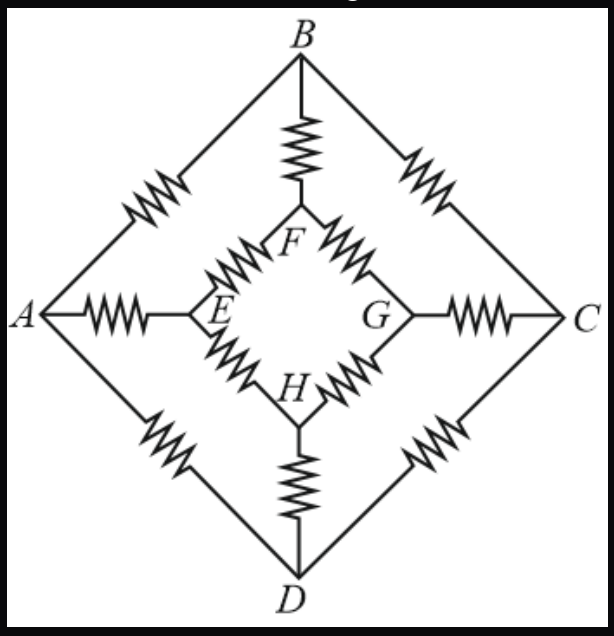
(B) v_e

(C) $2v_e$

(D) $\frac{v_e}{\sqrt{2}}$

Q.2

In the network shown in figure, each resistance is R.



The equivalent resistance between A and C is

(A)

$\frac{3R}{4}$

(B)

$\frac{5R}{6}$

(C)

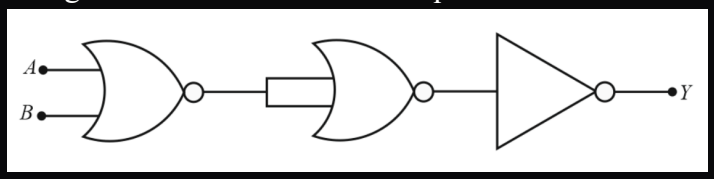
$\frac{7R}{12}$

(D)

$\frac{4R}{3}$

Q.3

The given electrical network is equivalent to



(A) AND gate

(B) OR gate

(C) NOR gate

(D) NOT gate

Q.4

In a cylinder-piston arrangement, the air is under a pressure P_1. A soap bubble of radius r lies inside the cylinder. Soap bubble has surface tension T. The radius of soap bubble is to be reduced to half. Find the pressure P_2 to which air should be compressed isothermally.

(A) $P_1 + \frac{4T}{r}$

(B) $4P_1 + \frac{12\,T}{r}$

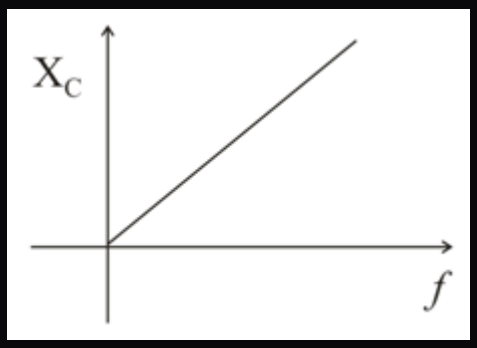
(C) $8P_1 + \frac{24\,T}{r}$

(D) $P_1 + \frac{2\,T}{r}$

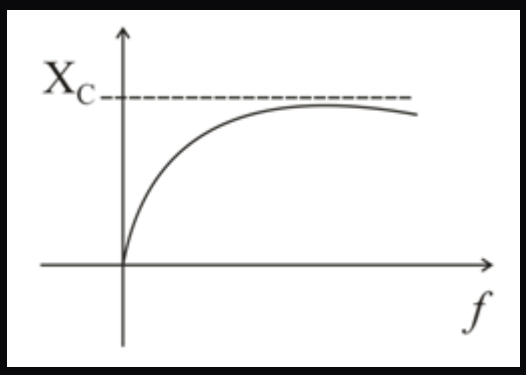
Q.5

Which of the following represents variation of capacitive reactance (X_C) and frequency (f) of the voltage source?

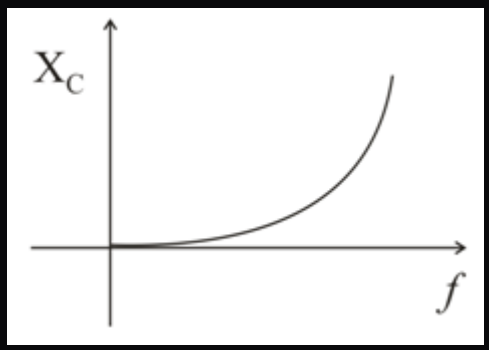
(A)



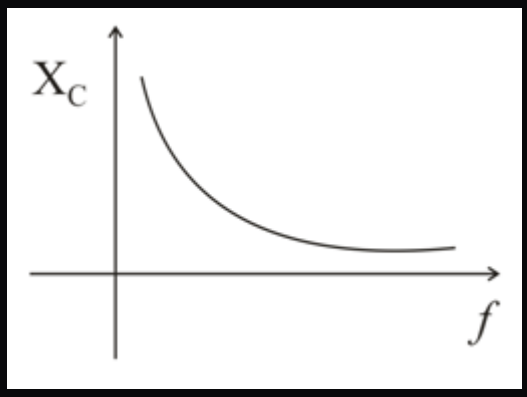
(B)



(C)



(D)



Q.6

Each atom of mass m of a monoatomic gas has got three degrees of freedom. The rms velocity of these atoms is v at temperature T. For a diatomic molecule of mass m and temperature T, which has got five degrees of freedom, rms velocity of molecule is,

(A) $\sqrt{\frac{5}{3}}v$

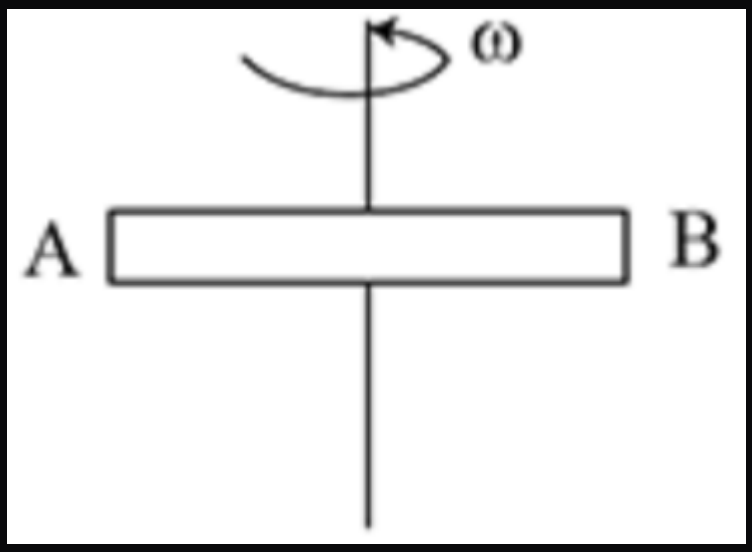
(B) $\sqrt{\frac{3}{5}}v$

(C) v

(D) $\sqrt{\frac{2}{5}}v$

Q.7

A non-conducting rod AB of length l has total charge q with constant charge and mass density. The rod is rotating about an axis passing through centre of mass with constant angular velocity ω as shown in the figure, the magnetic moment of the rod about the axis is :



(A) $\frac{q\omega l^2}{12}$

(B) $\frac{q\omega l^2}{3}$

(C) $\frac{q\omega l^2}{24}$

(D) $\frac{q\omega l^2}{6}$

Q.8

A wave pulse in a string is described by the equation $y_1 = \frac{5}{(3x-4t)^2+2}$ and another wave pulse in the same string is described by $y_2 = \frac{-5}{(3x+4t-6)^2+2}$. The values of y_1, y_2 and x are in metres and t is in seconds. Which of the following statement is correct?

(A) y_1 travels along $-x$ -direction and y_2 along $+x$ -direction

(B) Both y_1 and y_2 travel along $+x$ -direction

(C) At $x = 1m,\ y_1$ and y_2 always cancel

(D) At time $t = 1s,\ y_1$ and y_2 exactly cancel everywhere

Q.9

If the wavelength of the first line of the Balmer series of hydrogen is $6561 \mathring{A}$, find the wavelength of the second line of the series.

(A) $13122 \mathring{A}$

(B) $3280 \mathring{A}$

(C) $4860 \mathring{A}$

(D) $2187 \mathring{A}$

Q.10

A polarizer - analyser set is adjusted such that the intensity of light coming out of the analyser is just 36% of the original intensity after polarizer. Assuming that the polarizer - analyser set does not absorb any light, the angle by which the analyser needs to be rotated further, to reduce the output intensity to zero, is $\left(\sin^{-1}\left(\frac{3}{5}\right)=37^{\circ}\right)$

(A) 53°

(B) 37°

(C) 90°

(D) 45°

Q.11

A wheel of moment of inertia 2.5 Kg-m^2 has an initial angular velocity of 40 rads $^{-1}$. A constant torque of 10 Nm acts on the wheel. The time during which the wheel is accelerated to 60 rads^{-1} is

(A) 6 s

(B) 5 s

(C) 4.5 s

(D) 4 s

Q.12

A small conducting sphere of the radius r is lying concentrically inside a bigger hollow conducting sphere of radius R. The bigger and smaller sphere are charged with Q and $q(\mathrm{Q}>q)$ and are insulated from each other. The potential difference between the sphere will be

(A) $\frac{1}{4 \pi \varepsilon_0}\left(\frac{q}{r}-\frac{q}{R}\right)$

(B) $\frac{1}{4 \pi \varepsilon_0}\left(\frac{q}{R}-\frac{Q}{r}\right)$

(C) $\frac{1}{4 \pi \varepsilon_0}\left(\frac{q}{r}-\frac{Q}{R}\right)$

(D) $\frac{1}{4 \pi \varepsilon_0}\left(\frac{Q}{R}+\frac{q}{r}\right)$

Q.13

A system undergoes a reversible adiabatic process. The entropy of the system

(A) Remains constant

(B) May increase or may decrease

(C) Increases

(D) Decreases

Q.14

The question contains Statement 1 and Statement 2. Of the four choices given after the statements, choose the one that best describes the two statements.
Statement 1: Energy is released, when heavy nuclei undergo fission or light nuclei undergo fusion.
Statement 2: For heavy nuclei, the binding energy per nucleon increases with increasing Z, while for light nuclei, it decreases with increasing Z.

(A) Statement 1 is true, and Statement 2 is false.

(B) Statement 1 is false, and Statement 2 is true.

(C) Both the statements are true; Statement 2 is a correct explanation for Statement 1.

(D) Both the statements are true; Statement 2 is not a correct explanation for Statement 1.

Q.15

A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3 V s^{-1}. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor would be, respectively:

(A) zero, $60\ \mu A$

(B) $60\ \mu A$, $60\ \mu A$

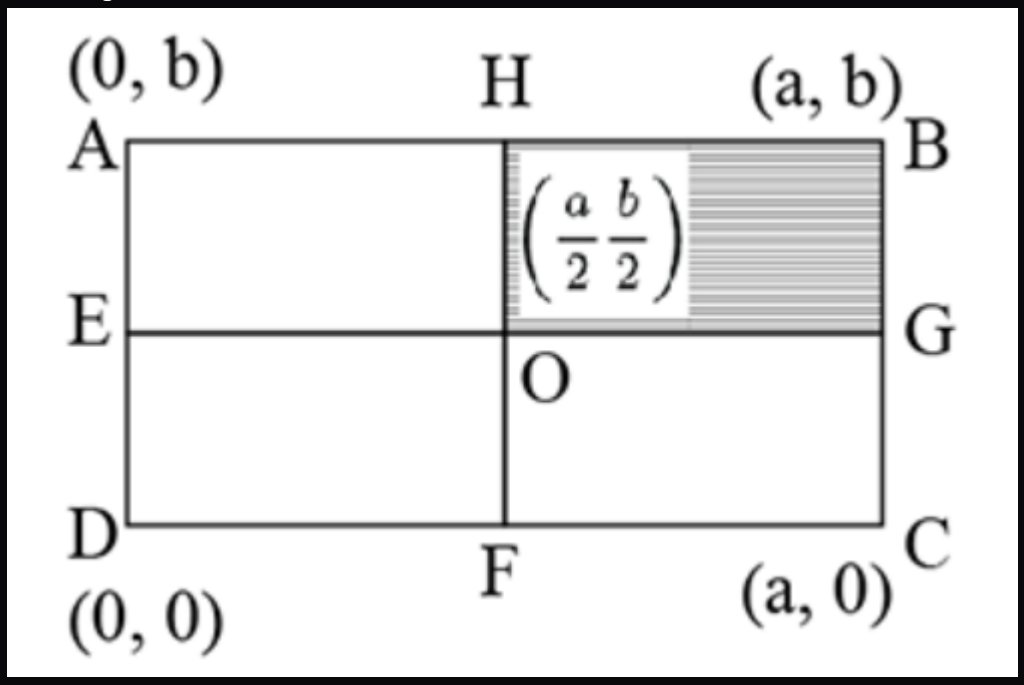
(C) $60\ \mu A$, zero

(D) zero, zero

Q.16

A uniform rectangular thin sheet $ABCD$ of mass M has length a and breadth b, as shown in the figure. If the

shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be:



(A) $\frac{5a}{12}, \frac{5\,b}{12}$

(B) $\frac{5a}{3}, \frac{5\,b}{3}$

(C) $\frac{2a}{3}, \frac{2\,b}{3}$

(D) $\frac{3a}{4}, \frac{3b}{4}$

Q.17

Energy (E), angular momentum (L) and universal gravitational constant (G) are chosen as fundamental quantities. The dimensions of universal gravitational constant in the dimensional formula of Planck's constant (h) is

(A) zero

(B) -1

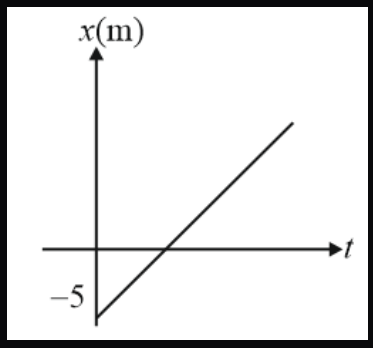
(C) $\frac{5}{3}$

(D) 1

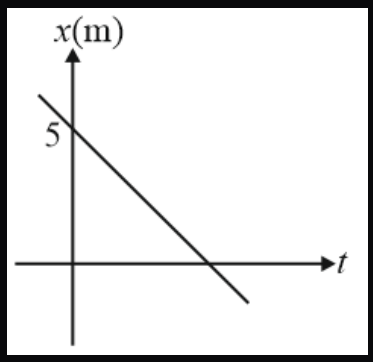
Q.18

A particle starts moving with constant velocity $v = 2\ \mathrm{m\ s^{-1}}$, from position $x = 5\ \mathrm{m}$. The position-time graph will be

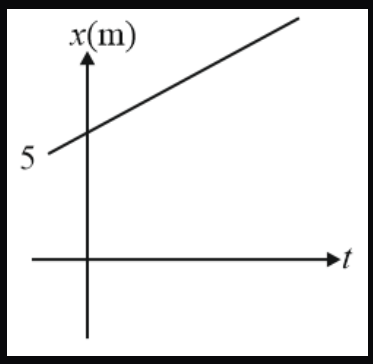
(A)



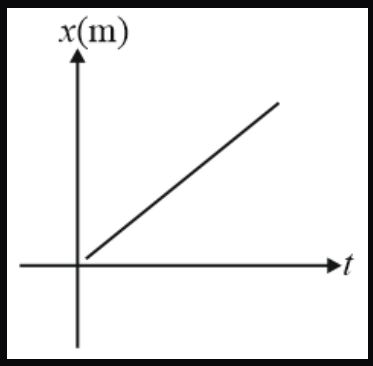
(B)



(C)



(D)



Q.19

There is some change in length when a $33000\ \mathrm{N}$ tensile force is applied on a steel rod of the area of cross-section $10^{-3}\ \mathrm{m}^2$. The change of temperature required to produce the same elongation, if the steel rod is heated, is (The modulus of elasticity is $3 \times 10^{11}\ \mathrm{N\ m^{-2}}$ and the coefficient of linear expansion of steel is $1.1 \times 10^{-5}\,^{\circ}\mathrm{C}^{-1}$)

(A) $20^{\circ}\mathrm{C}$

(B) $15^{\circ}\mathrm{C}$

(C) $10^{\circ}\mathrm{C}$

(D) $0^{\circ}\mathrm{C}$

Q.20

Two identical photocathodes receive light of frequencies f_1 and f_2. If the velocites of the photoelectrons (of mass m) coming out are with velocity v_1 and v_2 respectively, then

(A) $v_1^2 - v_2^2 = \frac{2h}{m}\left(f_1 - f_2\right)$

(B) $v_1 + v_2 = \left[\frac{2h}{m}\left(f_1 + f_2\right)\right]^{1/2}$

(C) $v_1^2 + v_2^2 = \frac{2h}{m}\left(f_1 + f_2\right)$

(D) $v_1 - v_2 = \left[\frac{2h}{m}(f_1 - f_2)\right]^{1/2}$

Q.21

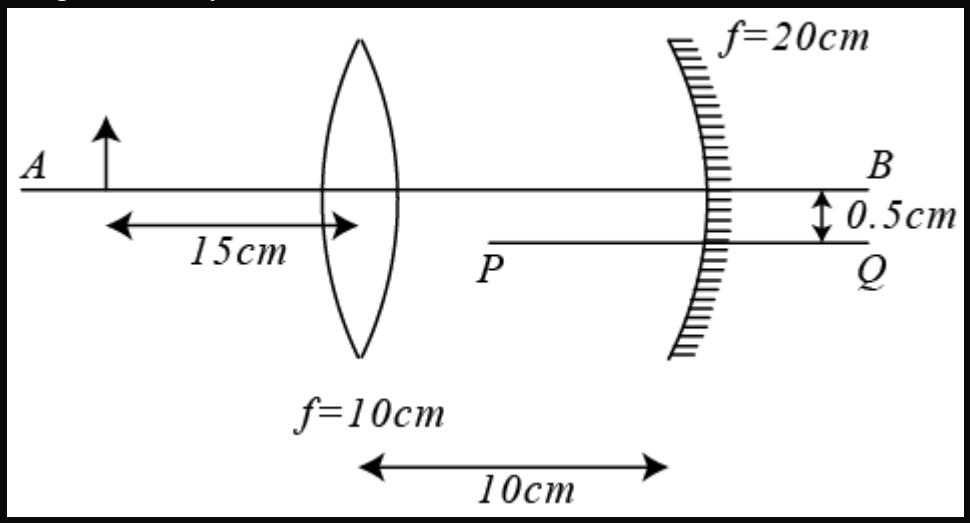
A physical quantity $X = \frac{A^2 B}{c^{1/3}\sqrt{D}}$ is calculated by using measured quantities A, B, C and D. If the errors in the measurement of A, B, C and D are $1\%, 2\%, 3\%$ and 4% respectively, find the percentage error in the measurement of X.

Q.22

Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the earth. If the escape speed on the surface of the earth is taken to be $11\ \mathrm{km/s}$, the escape speed on the surface of the planet in $\mathrm{km/s}$ will be

Q.23

A concave mirror of focal length $20\ \mathrm{cm}$ and a convex lens of focal length $10\ \mathrm{cm}$ are kept with their optic axes parallel but separated by $0.5\ \mathrm{cm}$ as shown in the figure. The distance between the lens and mirror is $10\ \mathrm{cm}$. An object of height $3\ \mathrm{mm}$ is placed on the optic axis of lens at a distance $15\ \mathrm{cm}$ from the lens. Find the length of the image formed by the mirror in mm.



Q.24

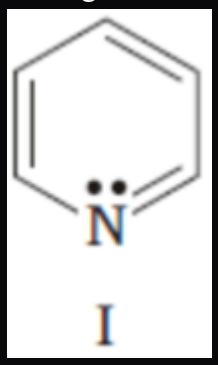
Two inductors L_1 (inductance 1 mH, internal resistance 3 Ω) and L_2 (inductance 2 mH, internal resistance 4 Ω), and a resistor R(resistance 12 Ω) are all connected in parallel across a 5 V battery. The circuit is switched on at time $t = 0$. The ratio of the maximum to minimum current $\left(\frac{I_{max}}{I_{min}}\right)$ drawn from the battery is:

Q.25

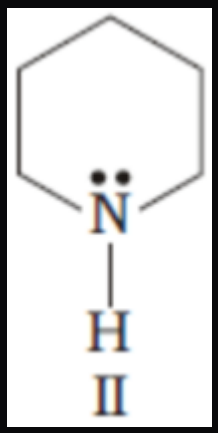
A ball is projected upwards from the top of the tower with a velocity 50 m s^{-1} making an angle 30° with the horizontal. The height of tower is 70 m . After how many seconds the ball will strike the ground? (take value of $g = 10\ m/s^2$)

Q.26

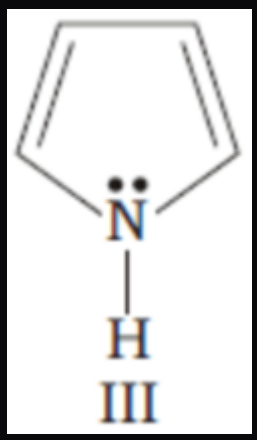
Arrange the following in correct decreasing order of basicity.



I



II



III

(A) $I > II > III$

(B) $II > II > I$

(C) $II > I > III$

(D) $I > III > II$

Q.27

In a flask containing 0.5 atm pressure of $A_2(g)$, some solid AB is added which undergoes dissociation according to $2AB(s) \rightleftharpoons A_2(g) + B_2(g), K_p = 0.06 \text{ atm}^2$

The total pressure (atm) at equilibrium will be

(A) 0.5 atm

(B) 0.2 atm

(C) 0.7 atm

(D) 1.4 atm

Q.28

When bromocyclohexane and sodium propynide are heated together, the product will be
I. $2-$cyclohexylpropyne
II. $1-$cyclohexylpropyne
III. Propyne
IV. Cyclohexene

(A) I and II

(B) II, III and IV

(C) I, II and II

(D) All four products are formed

Q.29

The solubility product values of barium sulphate and barium carbonate are 1.0×10^{-10} and 5.0×10^{-9} respectively. When the two salts are equilibrated in pure water the ratio of $\frac{[SO_4^{2-}]}{[CO_3^{2-}]}$ is

(A) 0.02

(B) 0.10

(C) 0.05

(D) 0.12

Q.30

When N_2 is converted into N_2^+ and O_2 is converted into O_2^+, then the bond strength, respectively.

(A) increases, increases

(B) decreases, decreases

(C) increases, decreases

(D) decreases, increases

Q.31

A $1.0M$ solution with respect to each of the metal halides AX_3, BX_2, CX_3 and DX_2 is electrolysed using platinum electrodes. If $E^\circ_{A^{3+}/A} = 1.50\ V, E^\circ_{B^{2+}/B} = 0.3\ V, E^\circ_{C^{3+}/C} = -0.74\ V, E^\circ_{D^{2+}/D} = -2.37\ V$. The correct sequence in which the various metals are deposited at the cathode is

(A) A, B, C, D

(B) A, B, C

(C) D, C, B, A

(D) C, B, A

Q.32

Statement 1: Friedel-Craft's reaction between benzene and acetic anhydride in the presence of anhydrous $AlCl_3$ yields acetophenone and not polysubstituted products.
Statement 2 : Acetophenone formed poisons the catalyst preventing further reaction.

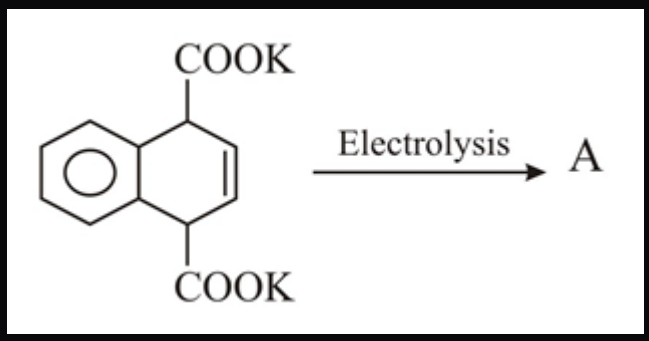
(A) Assertion is True, Reason is True; Reason is correct explanation for Assertion.

(B) Assertion is True, Reason is True; Reason is not correct explanation for Assertion.

(C) Assertion is True, Reason is False.

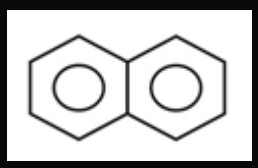
(D) Assertion is False, Reason is True.

Q.33

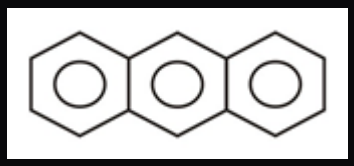


A is :

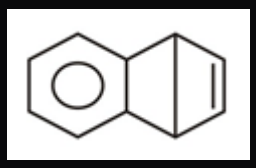
(A)



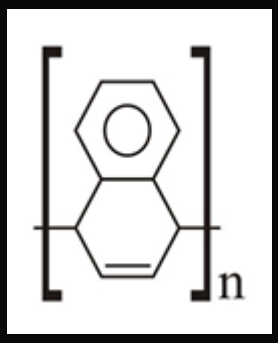
(B)



(C)



(D)



Q.34

Assertion: La (Z = 57) belongs to the d-block and is an element of group 3 of the periodic table.
Reason: All Lanthanoids belong to the group 3 of the periodic table.

(A) Both the assertion and reason statements are true and the reason is the correct explanation of the assertion

(B) Both assertion and reason statements are true but the reason is not the correct explanation of the assertion.

(C) The assertion statement is true but the reason statement is false

(D) Both the assertion and reason statements are false

Q.35

The vapour pressure of two pure liquids, A and B which form an ideal solution are 1000 and $1600\ \text{torr}$, respectively at $400\ \text{K}$. A liquid solution of A and B for which the mole fraction of A is 0.60 is contained in a cylinder by a piston on which the pressure can be varied. The solution is slowly vapourised at $400\ \text{K}$ by decreasing the applied pressure.
What is the composition of last droplet of liquid remaining in equilibrium with vapour?

(A) $X_A = 0.256, X_B = 0.74$

(B) $X_A = 0.40, X_B = 0.60$

(C) $X_A = 0.74, X_B = 0.256$

(D) $X_A = 0.50, X_B = 0.50$

Q.36

Match list I with list II and select the correct answer using the codes given below the lists.

List I (Pair of isomers)	List II (Type of isomerism)
(p) (I) $[Co(NH_3)_6][Cr(CN)_6]$, (II) $[Cr(NH_3)_6][Co(CN)_6]$	1. Ionization
(q) (III) $[PtCl_2(NH_3)_4]Br_2$, (IV) $[PtBr_2(NH_3)_4]Cl_2$	2. Hydrate
(r) (V) $[Co(SCN)(NH_3)_5]Cl_2$, (VI) $[Co(NCS)(NH_3)_5]Cl_2$	3. Coordination
(s) (VII) $[Cr(H_2O)_6]Cl_3$, (VIII) $[CrCl_2(H_2O)_4]Cl.2H_2O$	4. Geometrical
	5. Linkage isomerism

(A) p q r s
4 1 5 2

(B) p q r s
1 3 2 5

(C) p q r s
3 1 5 2

(D) p q r s
1 3 5 2

Q.37

Which of the following isoelectronic ions has lowest ionisation energy?

(A) Cl^-

(B) Ca^{2+}

(C) K^+

(D) S^{2-}

Q.38

In Duma's method for estimation of nitrogen $0.30\ g$ of organic compound gave $50\ cm^3$ of nitrogen gas collected at $300\ K$ and $715\ mm\ Hg$ pressure (Hint vapour pressure of water at $300\ K$ is $15\ mm\ Hg$) the value of percentage of nitrogen in the compound

(A) 17.46

(B) 28.56

(C) 28

(D) 15.68

Q.39

$2-$Bromobutane (A) is treated with NaI in the presence of dry acetone to give compound /B/. The compound /B/ is boiled with moist silver oxide to give compound /C/. Identify the compounds /B/ and /C/. How many optical isomers are possible for compound /C/?

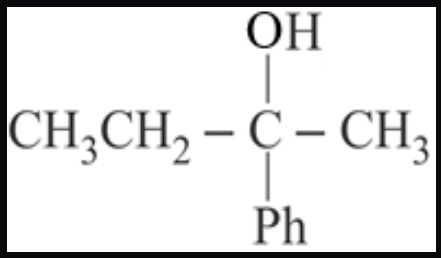
(A) $2-$Iodobutane, Butan$-2-$ol, 2 isomers

(B) $3-$Iodopentane, Pentan$-3-$ol, 4 isomers

(C) $2-$Iodobutane, Butan$-2-$ol, 4 isomers

(D) $3-$Iodopentane, Pentan$-3-$ol, 2 isomers

Q.40



cannot be prepared by:

(A) $HCHO + PhCH(CH_3)\,CH_2\,MgX$

(B) $PhCOCH_2\,CH_3 + CH_3\,MgX$

(C) $CH_3\,CH_2\,COCH_3 + PhMgX$

(D) $PhCOCH_3 + CH_3\,CH_2\,MgX$

Q.41

In a spontaneous process $\mathrm{X}+2\mathrm{Y} \rightarrow \mathrm{Z}$, the activation energy of forward and reverse reaction is the same, then

(A) $\Delta \mathrm{G}=0, \Delta \mathrm{S}=0, \Delta \mathrm{U}>0$

(B) $\Delta \mathrm{G}<0, \Delta \mathrm{S}<0, \Delta \mathrm{U}=0$

(C) $\Delta \mathrm{G}<0, \Delta \mathrm{S}>0, \Delta \mathrm{U}=0$

(D) $\Delta \mathrm{G}<0, \Delta \mathrm{S}>0, \Delta \mathrm{U}<0$

Q.42

The minimum and maximum number of electrons, which may have magnetic quantum number, $m=+1$ and spin quantum number, $s=-1/2$ in chromium (Cr) are respectively.

(A) $4,5$

(B) $0,1$

(C) $2,2$

(D) $2,3$

Q.43

Which of the following will not produce the benzyl amine?

(A) Benzamide $\xrightarrow{\mathrm{LiAlH_4}}$

(B) Phenyl nitrile $\xrightarrow{\mathrm{LiAlH_4}}$

(C) 2-phenylethyl isonitrile $\xrightarrow{\mathrm{LiAlH_4}}$

(D) 2-phenyl ethanamide $\xrightarrow{\mathrm{Br_2+KOH}}$

Q.44

Select the correct statement(s) among following.

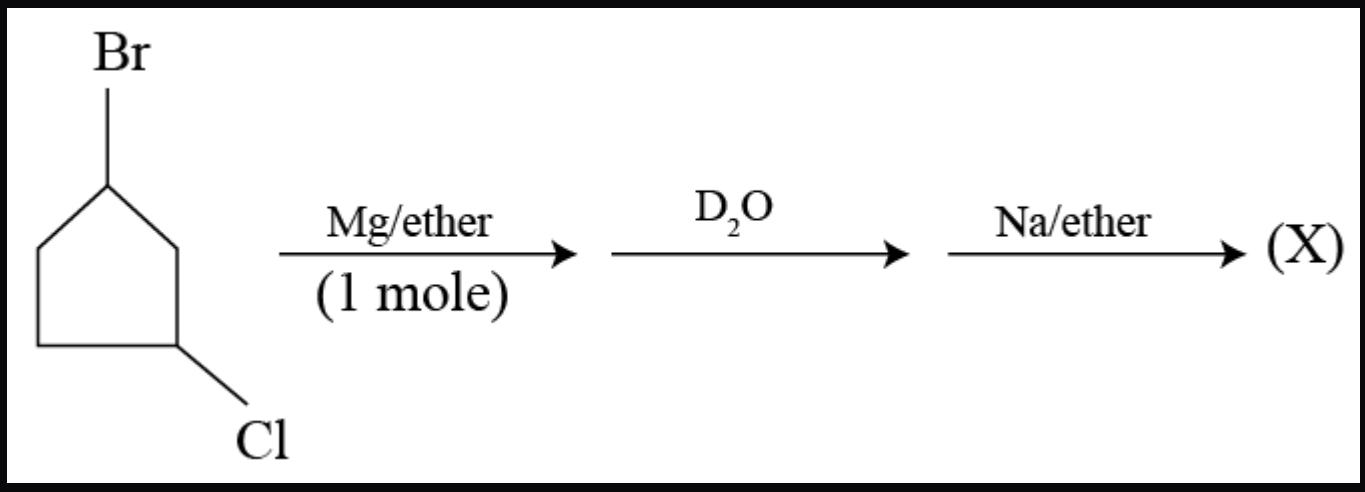
(A) Al forms AlF_6^{3-}, an octahedral complex

(B) Be and Al show diagonal relationship

(C) Be forms tetrahedral complex $\left[\mathrm{Be}\left(\mathrm{C}_2\mathrm{O}_4\right)_2\right]^{2-}$

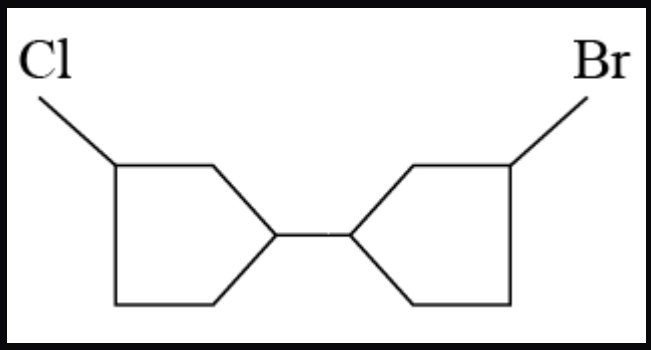
(D) All of the above

Q.45

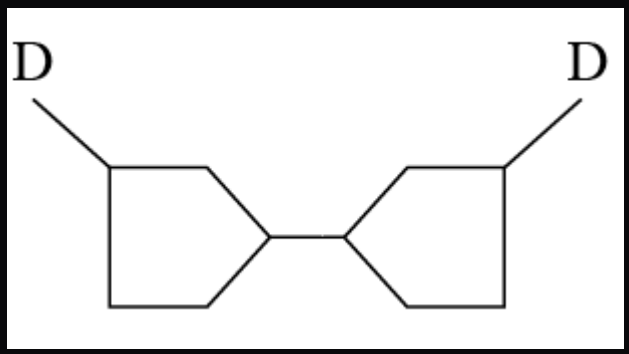


The compound (X) is

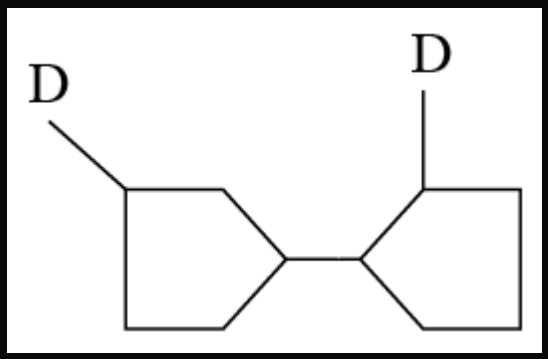
(A)



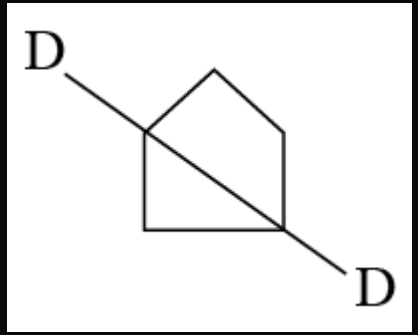
(B)



(C)



(D)



Q.46

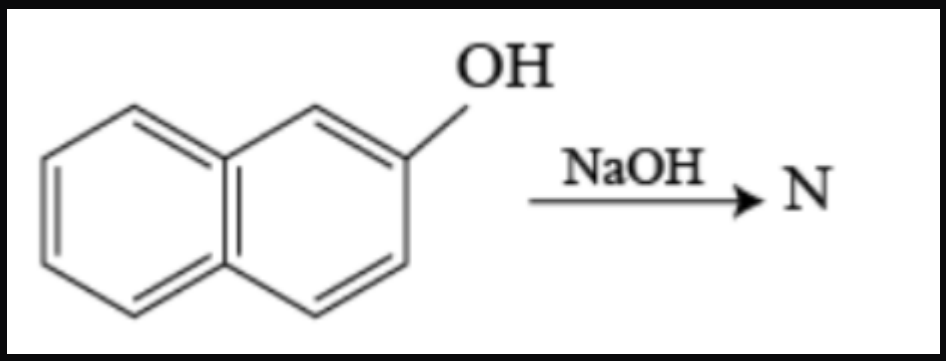
Concentration of NH_4Cl and NH_4OH in a buffer solution is in the ratio of $1:1$, K_b for NH_4OH is 10^{-10}. The pH of the buffer is

Q.47

The enthalpy of combustion of $H_2(g)$, to give $H_2O(g)$ is $-249\ kJ\ mol^{-1}$ and bond enthalpies of $H-H$ and $O=O$ are $433\ kJ\ mol^{-1}$ and $492\ kJ\ mol^{-1}$ respectively. The bond enthalpy of $O-H$ is _____ in KJ/mol

Q.48

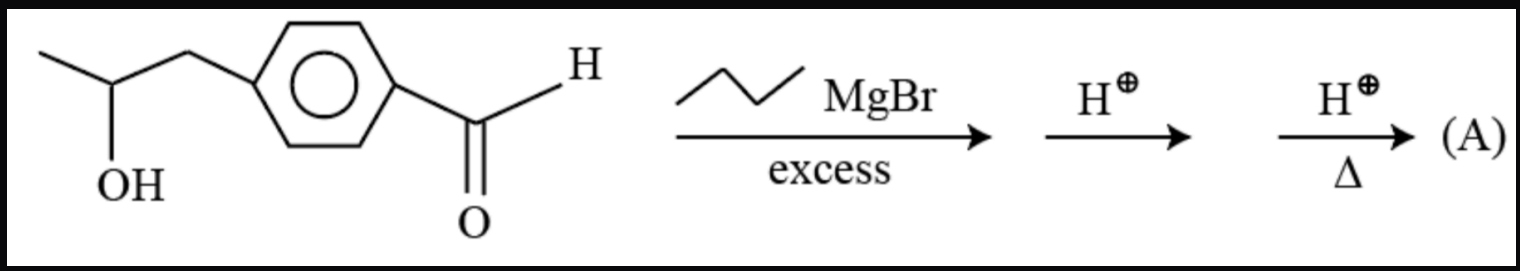
The number of resonance structures for N is



Q.49

In dilute aqueous H_2SO_4, the complex diaquodioxalatoferrate (II) is oxidised by MnO_4^-. For this reaction, the ratio of the rate of change of $[H^+]$ to the rate of change of $[MnO_4^-]$ is

Q.50



How many geometrical isomer of (A) is possible?

Q.51

A is one among the 8 horses in a race. A is to be ridden by one of the 3 jockeys P, Q, R. If P rides A all the horses are equally likely to win, if Q rides A his chances are doubled and if R rides A his chance are tripled. A die is thrown. If 1 or 2 or 3 appears then P rides A, if 4 or 5 appears then Q rides A, other-wise R rides A. Then the probability that A wins is

(A) $\frac{1}{12}$

(B) $\frac{3}{16}$

(C) $\frac{5}{24}$

(D) $\frac{7}{48}$

Q.52

If $f: R \rightarrow R$ is defined as $f(x)=\begin{cases} x+4, & x<-4 \\ 3x+2, & -4 \leq x<4, \\ x-4, & x \geq 4 \end{cases}$ then the value of $f(f(f(f(0))))+1$ is equal to:

(A) 0

(B) 1

(C) 2

(D) 4

Q.53

If A and B are square matrices of order 3 3. where $|A|=2$ and $|B|=3$, then $\left|\left(A^{-1}\right) \operatorname{adj}\left(B^{-1}\right) \cdot \operatorname{adj}\left(2 A^{-1}\right)\right|=$

(A) $\frac{8}{9}$

(B) $\frac{7}{9}$

(C) $\frac{2}{3}$

(D) $\frac{5}{3}$

Q.54

The equations $x^2+ax+12=0, x^2+bx+15=0$ and $x^2+(a+b)x+36=0$ have a common positive root then the value of $a-b$ is

(A) 1

(B) 2

(C) 3

(D) 4

Q.55

Let n be the coefficient of x^6 in $\left(1-x^3\right)^5\left(1+x^2\right)^4\left(1+x^4\right)^8$, when expanded in powers of ' x '. The value of $\left[\frac{n}{19}\right]$ is (where [.] denotes greatest integer function)

(A) 1

(B) 3

(C) 2

(D) 5

Q.56

In the xy-plane, how many straight lines whose x-intercept is a prime number and whose y-intercept is a positive integer pass through the point $(4,3)$?

(A) 1

(B) 0

(C) 3

(D) 2

Q.57

The number of complex numbers z such that $\arg(z+3i)=\frac{\pi}{4}$ and $\arg(z-2)=\frac{\pi}{3}$ is

(A) 0

(B) 1

(C) 2

(D) 3

Q.58

If N is the number of positive integral solutions of $x_1x_2x_3x_4=770$, then $\mathrm{N}=$

(A) 256

(B) 729

(C) 900

(D) 770

Q.59

If equation $8\mathrm{x}^3-6\mathrm{x}+1=0$ has roots α,β,γ where $\alpha<\beta<\gamma$, then $\{$ where [ ] denotes greatest integer function$\}$ Mark incorrect option

(A) $[|\alpha|+|\beta|+|\gamma|]=1$

(B) $[\alpha]+[\beta]+[\gamma]=-1$

(C) $18\beta<\pi$

(D) $\sin^{-1}\gamma=\frac{7\pi}{18}$

Q.60

In a quadrilateral $\mathrm{ABCD}, \overrightarrow{AC}$ is the bisector of $\overrightarrow{AB}$ and $\overrightarrow{AD}$, angle between $\overrightarrow{AB}$ and $\overrightarrow{AD}$ is $\frac{2\pi}{3}, 15|\overrightarrow{AC}|=3|\overrightarrow{AB}|=5|\overrightarrow{AD}|$. Then the angle between $\overrightarrow{BA}$ and $\overrightarrow{CD}$ is

(A) $\cos^{-1}\frac{\sqrt{14}}{7\sqrt{2}}$

(B) $\cos^{-1}\frac{\sqrt{21}}{7\sqrt{3}}$

(C) $\cos^{-1}\frac{2}{\sqrt{7}}$

(D) $\cos^{-1}\frac{2\sqrt{7}}{14}$

Q.61

Let $f(x)=[x]\sin\frac{\pi}{[x+1]}$, The points of discontinuity are

(A) Z

(B) $Z-\{0\}$

(C) $Z-\{1\}$

(D) $Z-\{-1\}$

Q.62

Let $f(x)$ be an even function, $I_1=\int_0^{\frac{\pi}{2}}f(\cos 2x)\cos x dx, I_2=\int_0^{\frac{\pi}{4}}f(\sin 2x)\cos x dx$. Then $\frac{I_1}{I_2}=$

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{2}}$

(D) $\sqrt{2}$

Q.63

If the eccentricity of the hyperbola $x^2-y^2\sec^2\theta=5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2\sec^2\theta+y^2=25$, then smallest positive value of θ is $\frac{\pi}{P}$, where twice the value of 'p' is

(A) 8

(B) 12

(C) 16

(D) 24

Q.64

The area bounded by the curve $\sqrt{x}+\sqrt{y}=1$ and the coordinate axes is

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $\frac{1}{6}$

Q.65

The mean and standard deviation of 10 observations $x_1, x_2, x_3 \ldots x_{10}$ are $\bar{x}$ and σ respectively. Let 10 is added to $x_1, x_2 \ldots x_9$ and 90 is subtracted from x_{10}. If still, the standard deviation is the same, then $x_{10} - \bar{x}$ is equal to

(A) 35

(B) 45

(C) 55

(D) 50

Q.66

Let $\vec{x} = \vec{a} + \vec{b}, \vec{y} = 2\vec{a} - \vec{b}$, then the point of intersection of straight lines $\vec{r} \times \vec{x} = \vec{y} \times \vec{x}, \vec{r} \times \vec{y} = \vec{x} \times \vec{y}$ is

(A) $2\vec{b}$

(B) $3\vec{b}$

(C) $3\vec{a}$

(D) $2\vec{a}$

Q.67

If $f(x) = e + (1-x)\ln\frac{x}{e} + \int_1^x f(t)dt$, then $f(e) =$

(A) e^e

(B) $e^e - 1$

(C) $e^e + 1$

(D) e^{e+1}

Q.68

The solution set of the inequality $\left(\text{cosec}^{-1} x\right)^2 - 2\left(\text{cosec}^{-1} x\right) \geq \frac{\pi}{6}\,\text{cosec}^{-1} x - \frac{\pi}{3}$ is $(-\infty, a] \cup [b, \infty)$, then $(a+b)$ is equal to

(A) 0

(B) 2

(C) -3

(D) 1

Q.69

The sum of four numbers in A.P. is 48, and the product of the extremes is to the product of the two middle terms as 27 to 35. The largest term of the A. P. is

(A) 10

(B) 12

(C) 14

(D) 18

Q.70

If the function $\mathrm{f(x)}$ and $\mathrm{g(x)}$ are defined on $R \to R$ such that $f(x) = \begin{cases} \mathrm{x+3}, & \mathrm{x} \in \text{ rational} \\ \mathrm{4x}, & \mathrm{x} \in \text{ irrational} \end{cases}$ and $g(x) = \begin{cases} x + \sqrt{5}, & x \in \text{ irrational} \\ -x, & x \in \text{ rational} \end{cases}$ then $(f-g)(x)$ is

(A) one-one and onto

(B) neither one-one nor onto

(C) one-one but not onto

(D) onto but not one-one

Q.71

When $(3053)^{456} - (2417)^{333}$ is divided by 9 the remainder is ____.

Q.72

If number of arrangements of letters of the word "DHARAMSHALA" taken all at a time so that no two alike letters appear together is $\left(4^{\mathrm{a}}.5^{\mathrm{b}} \cdot 6^{\mathrm{c}} \cdot 7^{\mathrm{d}}\right)$, (where $\mathrm{a, b, c, d} \in \mathrm{N}$), then $\mathrm{a+b+c+d}$ is equal to

Q.73

Find the shortest distance between the lines
$\frac{x+2}{2} = \frac{y+6}{3} = \frac{z-34}{-10}, \frac{x+6}{4} = \frac{y-7}{-3} = \frac{z-7}{-2}$.

Q.74

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $P = \begin{bmatrix} p \\ q \end{bmatrix}$. Such that $AP = P$ and $a + d = 5050$. Find the value of (ad - bc-5040).

Q.75

If x is a complex root of the equation $\begin{vmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{vmatrix} + \begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix} = 0$ then $x^{2011} + \frac{1}{x^{2011}} =$

Answers & Solutions

Q.1 Answer:

—

Solution:

From mechanical energy conservation,

$\left[\frac{-GmM_e}{R_e}+\frac{1}{2}mv'^2\right]i$

$=\left[-\frac{GmM_e}{r}+(KE)_f\right]$

$\frac{-GmM_e}{R_e}+\frac{1}{2}mv'^2=\frac{-GmM_e\times 2}{3R_e}+0\left[r=\frac{3R_e}{2}\right]$

$\Rightarrow v'^2=\frac{2GM_e}{3R_e}$

or $v'=\frac{v_e}{\sqrt{3}}\quad\left[\because v_e=\sqrt{\frac{2GM_e}{R_e}}\right]$

Q.2 Answer:

$\frac{3\mathrm{R}}{4}$

Solution:

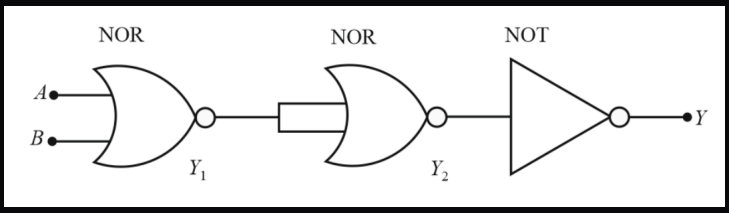
If we connect the battery across A **and** C**, then no current will flow in the resistances between** B **and** F **and between** H **and** D**. So they can be removed. Thus,**

$\mathrm{R_{eq}}=\frac{3\mathrm{R}}{4}$

Q.3 Answer:

NOR gate

Solution:



Output at NOR gate $Y_1=\overline{A+B}$

The output of first NOR gate is input to the second NOR gate, so the output of second NOR gate is

$Y_2=\overline{Y_1+Y_1}=\overline{Y_1}=\overline{\overline{A+B}}=A+B$

The output of NOT gate is $Y=\overline{Y_2}=\overline{A+B}$

The given electrical network becomes equivalent to the NOR gate.

Q.4 Answer:

$8\mathrm{P}_1+\frac{24\,\mathrm{T}}{\mathrm{r}}$

Solution:

Using $P_iV_i=P_fV_f$ **(for the air bubble)**

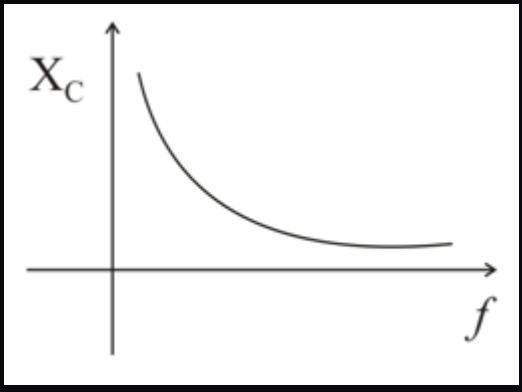
we have

$\left(\mathrm{P}_1+\frac{4\,\mathrm{T}}{\mathrm{r}}\right)\left(\frac{4}{3}\pi\mathrm{r}^3\right)=\left(\mathrm{P}_2+\frac{4\,\mathrm{T}}{\mathrm{r}/2}\right)\left[\frac{4}{3}\pi\left(\frac{\mathrm{r}}{2}\right)^3\right]$

Solving this, we get

$\mathrm{P_2 = 8P_1 + \frac{24\,T}{r}}$

Q.5 Answer:



Solution:

The formula of capacitance reactance in terms of angular frequency and capacity,

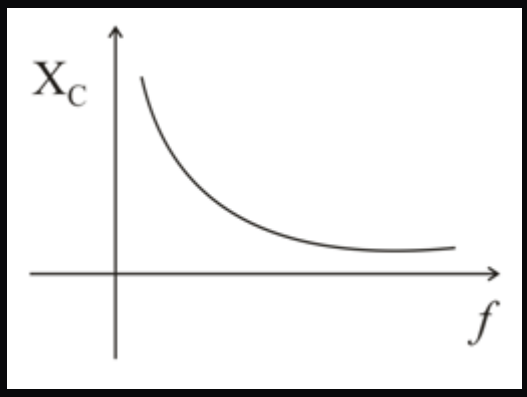
$X_{\mathrm{C}} = \frac{1}{\omega C}$

here the value of angular frequency, $\omega = 2\pi f$

$X_{\mathrm{C}} = \frac{1}{2\pi f C}$

$X_{\mathrm{C}} \propto \frac{1}{f}$

so the value of X_{C} is inversely proportional to frequency,



Q.6 Answer:

v

Solution:

The rms velocity of a gas molecule is,

$v_{\mathrm{rms}} = \sqrt{\frac{3RT}{M}}$.

Therefore, the rms velocity is independent of the degree of freedom.

Q.7 Answer:

$\frac{q\omega l^2}{24}$

Solution:

$$\frac{M}{L}=\frac{q}{2m}$$
$$M=\frac{q}{2m}\times I\omega=\frac{q}{2m}\times\frac{ml^2}{12}\times\omega=\frac{q\omega l^2}{24}$$

Q.8 Answer:

At $x=1m,\ y_1$ and y_2 always cancel

Solution:

$\because$ **Given,**

$y_1=\frac{5}{(3x-4t)^2+2},\ y_2=\frac{-5}{(3x+4t-6)^2+2}$

According option (d) **, at** $x=1$

$y_1=\frac{5}{(3-4t)^2+2}$

$y_2=\frac{-5}{(3+4t-6)^2+2}$

$=\frac{-5}{(3-4t)^2+2}$

Both wave pulse equation are existing in same string therefore resultant equation of wave pulse.

$y=y_1+y_2=0$

Hence, option At $x=1m,\ y_1$ **and** y_2 **always cancel is correct.**

Q.9 Answer:

$4860\mathring{A}$

Solution:

The wavelength of spectral line in Balmer series is given by $\frac{1}{\lambda}=\mathrm{R}\left[\frac{1}{2^2}-\frac{1}{\mathrm{n}^2}\right]$

For first line of Balmer series, $n=3$

$$\Rightarrow\frac{1}{\lambda_1}=\mathrm{R}\left[\frac{1}{2^2}-\frac{1}{3^2}\right]=\frac{5\mathrm{R}}{36};\ \text{For second line n}=4$$
$$\Rightarrow\frac{1}{\lambda_2}=\mathrm{R}\left[\frac{1}{2^2}-\frac{1}{4^2}\right]=\frac{3\mathrm{R}}{16}$$
$$\therefore\frac{\lambda_2}{\lambda_1}=\frac{20}{27}\Rightarrow\lambda_2=\frac{20}{27}\times6561=4860$$

Q.10 Answer:

37°

Solution:

$I = I_0 \cos^2\theta$

$0.36 I_0 = I_0 \cos^2\theta$

$\cos(\theta) = \frac{3}{5}$

$\Rightarrow \theta = 53^\circ$

angle rotated should be $= 90^\circ - 53^\circ = 37^\circ$

Q.11 Answer:

5 s

Solution:

Given, $\text{MI} = 2.5\ \text{kgm}^{-2}$

$W = 40\ \text{rad s}^{-1}$

$T = 10\ \text{Nm}$

As $T = I\alpha$

$10 = 2.5\alpha$

$\alpha = 4\ \text{rad s}^{-2}$

Now, $\omega = \omega_0 + \alpha t$

$60 = 40 + 4 \times t$

$20 = 4t$

$t = 5\ s$

Q.12 Answer:

$\frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r} - \frac{q}{R}\right)$

Solution:

Potential of the inner sphere

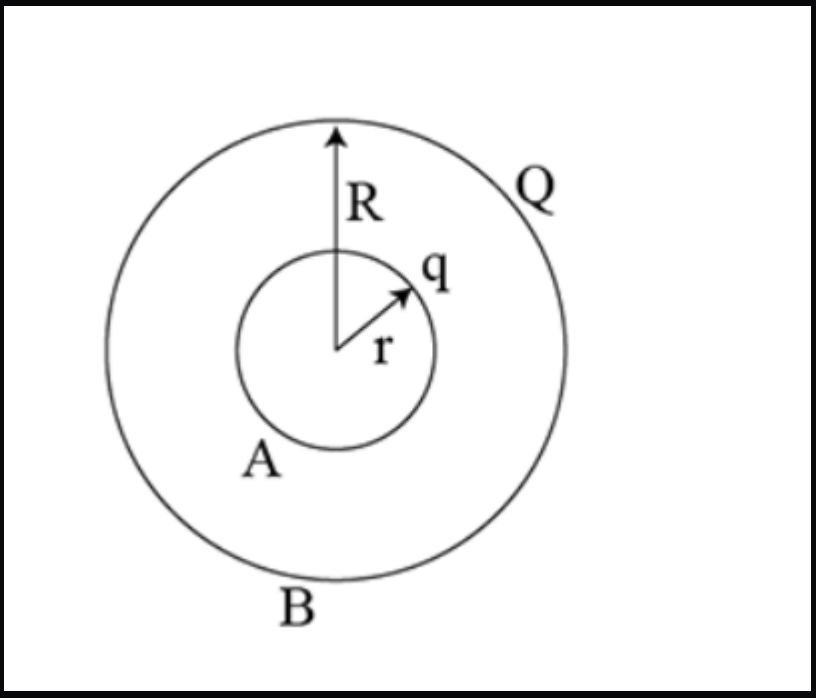
$V_{\text{A}} = \frac{kq}{r} + \frac{kQ}{R}$

Potential of the outer sphere

$$V_B = \frac{kq}{R} + \frac{kQ}{R}$$

$$\therefore V_A - V_B = \frac{kq}{r} - \frac{kq}{R}$$

$$= \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r} - \frac{q}{R}\right)$$



Q.13 Answer:

Remains constant

Solution:

In general, the temperature of the system may change during a process. If the process is reversible, the change in entropy is defined as

$S_f - S_i = \int_i^f \frac{\Delta Q}{T}$

In an adiabatic reversible process, no heat is given to the system.

Therefore, entropy of the system remains constant.

Q.14 Answer:

Statement 1 is true, and Statement 2 is false.

Solution:

Statement 1 is correct, while Statement 2 is false. Statement 2 is false for the reason that for heavy nuclei, the binding energy per nucleon decreases with increasing Z; while for light nuclei, it increases with increasing Z. Therefore, the option (Statement 1 is true, and Statement 2 is false) is correct.

Q.15 Answer:

60 μA, 60 μA

Solution:

$q = CV$

$\frac{dq}{dt} = C\frac{dv}{dt}$

$= 20 \times 10^{-6} \times 3$

Conduction current $= 60\,\mu A$

Displacement current $= \varepsilon_0 \frac{d\phi_e}{dt}$

$= \varepsilon_0 \frac{A}{d} \frac{dV}{dt}$

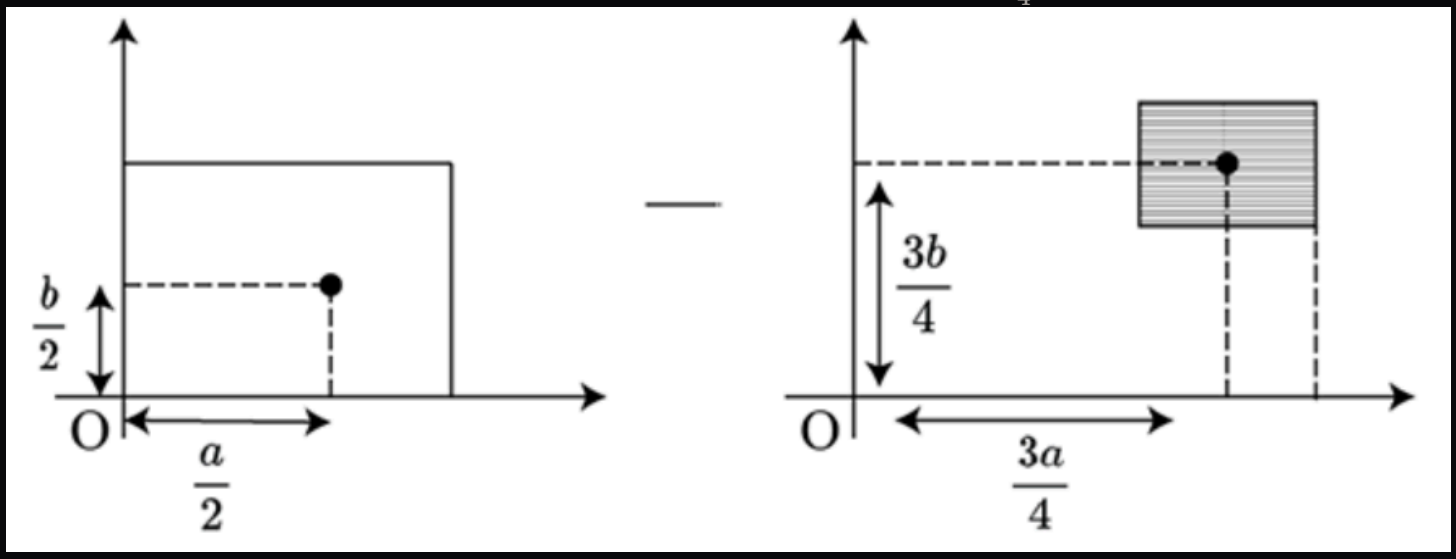
$= 60\,\mu A$

Q.16 Answer:

$\frac{5a}{12}, \frac{5b}{12}$

Solution:

If the total mass of object mass ' M **' then the mass removed part will be** $\frac{M}{4}$.



$$X_{CM} = \frac{M \cdot \frac{a}{2} - \frac{M}{4}\left(\frac{3a}{4}\right)}{M - \frac{M}{4}} = \frac{\frac{Ma}{2} - \frac{3Mb}{16}}{\frac{3M}{4}} = \frac{5}{12}a$$

$$Y_{CM} = \frac{M \cdot \frac{b}{2} - \frac{M}{4}\left(\frac{3b}{4}\right)}{M - \frac{M}{4}} = \frac{\frac{Mb}{2} - \frac{3Ma}{10}}{\frac{3M}{4}} = \frac{5}{12}b$$

Q.17 Answer:

zero

Solution:

$h = G^x L^y E^z$

$[M^1 L^2 T^{-1}] = [M^{-1} L^3 T^{-2}]^x [M^1 L^2 T^{-1}]^y [M^1 L^2 T^{-2}]^z$

Comparing the power, we get

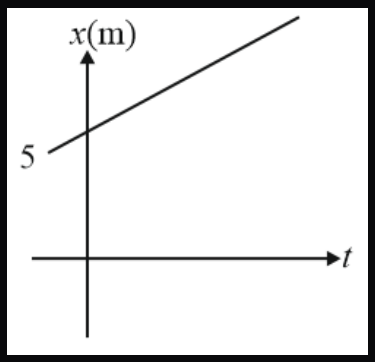
$1 = -x + y + z$ ... (i)

$2 = 3x + 2y + 2z$ ... (ii)

$-1 = -2x - y - 2z$ ... (iii)

On solving Eq. (i), (ii) and (iii), we get x = 0

Q.18 Answer:



Solution:

Here the particle is moving with a constant velocity, so its acceleration $a = 0\ \mathrm{m\ s^{-2}}$, initial velocity $u = 2\ \mathrm{m\ s^{-1}}$. Initial position of the particle is $x_0 = 5\ \mathrm{m}$. Position at time t is given by equation of motion,

$x = x_0 + ut + \frac{1}{2}at^2$, **$\because a = 0$,**

$\Rightarrow x = x_0 + ut = 5 + 2t$

$\therefore x = 2t + 5 \ \ldots (1)$

This equation is of the form $y = mx + c$ where m is the slope of the straight line and c is the y-intercept. On comparison with equation (1), it is found that $m = 2$ and $c = 5$.

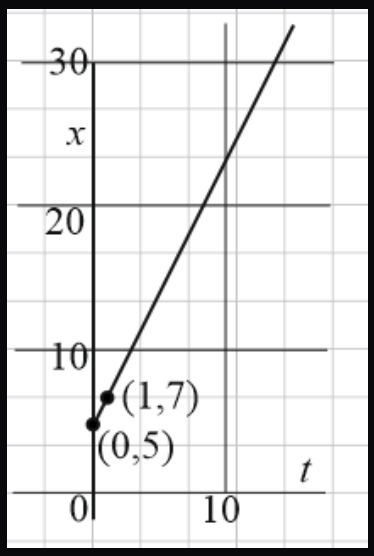
Since $m > 0$, the straight line corresponding to (1) will make an angle with the positive x-axis which is less than 90°.

At $t = 0$ s, $x = 2(0) + 5 = 5 \Rightarrow (0, 5)$ will lie on the straight line.

At $t = 1$ s, $\mathrm{x} = 2(1) + 5 = 7 \Rightarrow (1, 7)$ will lie on the straight line.

Another thing is that time $t \geq 0$.

Joining these two points will give the required straight line but this line will not be in the region where t is less than 0.



Q.19 Answer:

$10^\circ \mathrm{C}$

Solution:

Modulus of elasticity $= \frac{\text{Force}}{\text{Area}} \times \frac{l}{\Delta l}$

$3 \times 10^{11} = \frac{33000}{10^{-3}} \times \frac{l}{\Delta l}$

$\frac{\Delta l}{l} = \frac{33000}{10^{-3}} \times \frac{1}{3 \times 10^{11}}$

$= 11 \times 10^{-5}$

Change in length, $\frac{\Delta l}{l} = \alpha \Delta T$

$11 \times 10^{-5} = 1.1 \times 10^{-5} \times \Delta T$

$\Rightarrow \Delta T = 10 \text{ K or } 10^{\circ}\text{C}$

Q.20 Answer:

$v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

Solution:

For one photocathode

$hf_1 - W = \frac{1}{2}mv_1^2 ....(\text{i})$

For another photo cathode

$hf_2 - W = \frac{1}{2}mv_2^2 .... (\text{ii})$

Subtracting (ii) from (i) we get

$(hf_1 - W) - (hf_2 - W) = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2$

$\therefore h(f_1 - f_2) = \frac{m}{2}(v_1^2 - v_2^2)$

$\therefore v_1^2 - v_2^2 = \frac{2h}{m}(f_1 - f_2)$

Q.21 Answer:

—

Solution:

Given $X = \frac{A^2B}{c^{1/3}\sqrt{D}}$

$\frac{\Delta X}{X} = 2\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{1}{3}\frac{\Delta C}{C} + \frac{1}{2}\frac{\Delta D}{D}$

$= 2 \times 1\% + 2\% + \frac{1}{3} \times 3\% + \frac{1}{2} \times 4\%$

$= 2\% + 2\% + 1\% + 2\%$

$= 7\%$

Q.22 Answer:

—

Solution:

$$g = \frac{GM}{R^2} = \frac{G}{R^2}\frac{4}{3}\pi R^3\rho = \frac{4}{3}\pi GR\rho \text{ i.e. } g \propto R\rho$$

$$\frac{g_P}{g_E} = \frac{\sqrt{6}g_E}{11g_E} = \frac{\sqrt{6}}{11} \text{ and } \frac{\rho_E}{\rho_P} = \frac{3}{2} \text{ and } \frac{g_P}{g_E} = \frac{R_P\rho_P}{R_E\rho_E}$$

$$\therefore \frac{R_P}{R_E} = \frac{g_P\rho_E}{g_E\rho_P} = \frac{\sqrt{6}}{11} \times \frac{3}{2}$$

$\therefore$ Escape velocity $v = \sqrt{2gR}$

$$\therefore \frac{v_P}{v_E} = \sqrt{\frac{g_P}{g_E} \times \frac{R_P}{R_E}} = \sqrt{\frac{\sqrt{6}}{11} \times \frac{\sqrt{6}}{11} \times \frac{3}{2}} = \frac{3}{11}$$

$$\therefore v_P = \frac{3}{11} \times v_E = \frac{3}{11} \times 11 = 3 \text{ km s}^{-1}$$

Q.23 Answer:

—

Solution:

For image formed by the lens $\frac{1}{v_1} - \frac{1}{-15} = \frac{1}{+10}$

$\Rightarrow v_1 = +30$ cm

i.e. 20 cm **behind mirror For mirror** $\frac{1}{v_2} + \frac{1}{20} = \frac{1}{-20}$

$\Rightarrow v_2 = -10$ cm

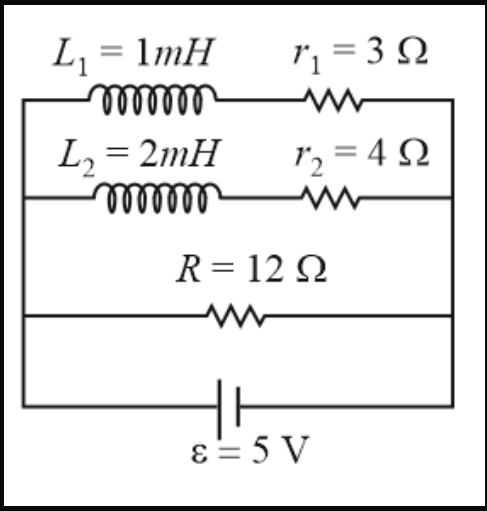
Overall magnification $= \left(\frac{30}{-15}\right) \times \left(\frac{10}{20}\right) = -1$

Length of image $= 1 \times 3 = 3$ mm

Q.24 Answer:

—

Solution:



Initially, the inductances will offer infinite resistance and the current will be maximum.

$I_{max} = \frac{\varepsilon}{R} = \frac{5}{12}$ A **(Initially at** $t = 0$**)**

Upon reaching steady state, the inductances will act as pure resistors and hence the current will be minimum.

$I_{min} = \frac{\varepsilon}{R_{eq}} = \varepsilon\left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{R}\right)$ (Finally in steady state)

$= 5\left(\frac{1}{3} + \frac{1}{4} + \frac{1}{12}\right)$

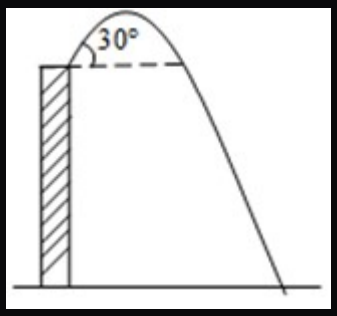
$= \frac{10}{3} A$

$\frac{I_{max}}{I_{min}} = 8$

Q.25 Answer:

—

Solution:



$\overline{\mathrm{Y}} = \overline{\mathrm{v}}_{\mathrm{y}}\mathrm{t} + \frac{1}{2}\overline{\mathrm{a}}_{\mathrm{y}}\mathrm{t}^2$

$70 = (-50\sin 30^\circ)\mathrm{t} + \frac{1}{2} \times 10 \times \mathrm{t}^2$

$70 = -25\mathrm{t} + 5\mathrm{t}^2$

or $\mathrm{t}^2 - 5\mathrm{t} - 14 = 0$

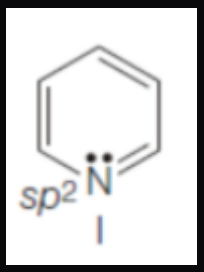
$\mathrm{t} = 7$ s ($-$ve time is not possible)

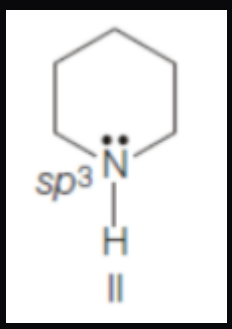
Q.26 Answer:

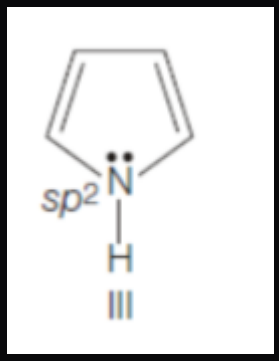
$II > I > III$

Solution:

Hybridisation of N-atom in above compounds are sp^2, sp^3 and sp^2 respectively.

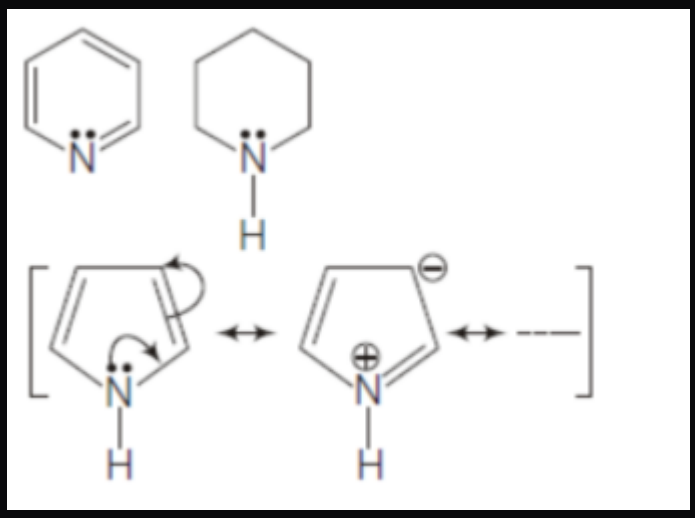






Greater the $s-$character, more will be electronegativity of N-atom and lesser will be its basicity. On this basis I is less basic than II

Conjugation If lone pairs of electron of N are involved in conjugation causes decrease in basicity of compound due to lesser availability of lone pair for donation to show basic nature.



Lone pair involved in formation of aromatic sextet of 6π-electron (least basic)

Q.27 Answer:

0.7 atm

Solution:

$2\mathrm{AB}(\mathrm{s}) \rightleftharpoons \mathrm{A}_2(\mathrm{~g})+\mathrm{B}_2(\mathrm{~g})$
$(0.5+p_0)\ \mathrm{p}_0$
$K_p=p_{A_2} \times p_{B_2}$
$0.06=(0.5+p_0)\, p_0$
$p_0^2+0.5 p_0-0.06=0$
$p_0=0.1$
$p_{\text{total }}=p_{A_2}+p_{\mathrm{B}_2}=(0.5+p_0)+p_0$
$=0.6+0.1=0.7 \text{ atm}$

Q.28 Answer:

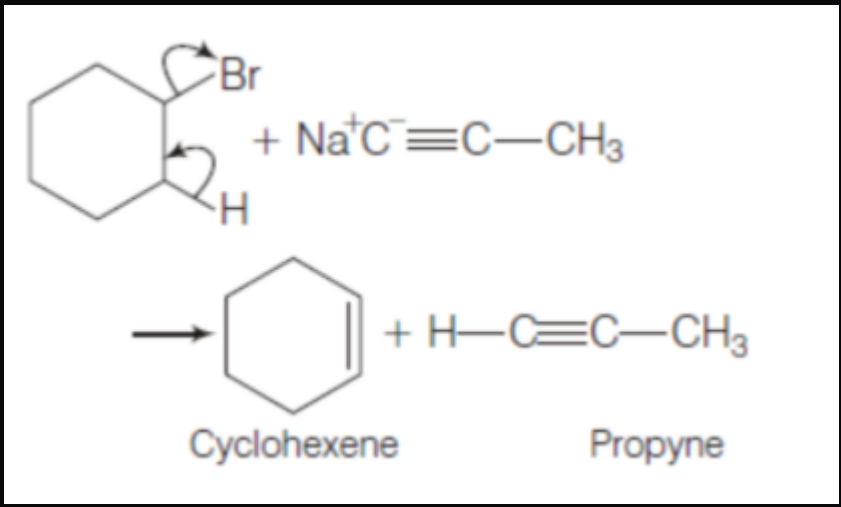
II, III and IV

Solution:

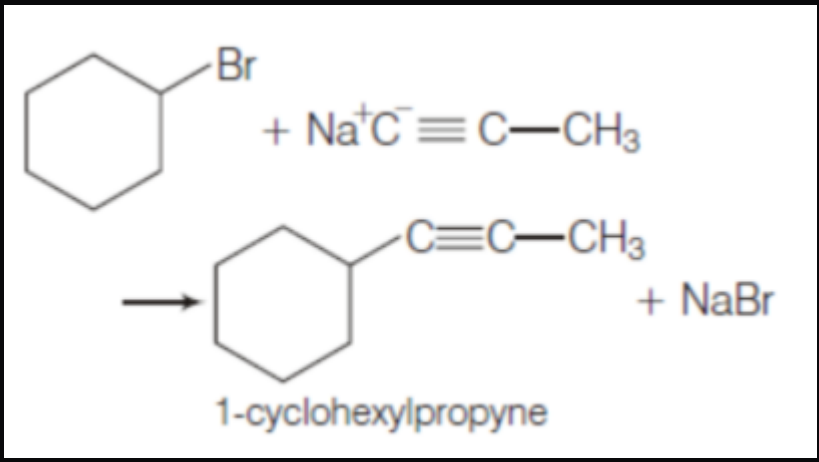
2−cyclohexylpropyne is wrong name Compound with such name does not exist.

The other three compounds are formed as

In elimination



In substitution,



The yield of substituting product will be greater than that of elimination products.

Q.29 Answer:

0. 02

Solution:

$$K_{sp_1} = [Ba^{2+}][SO_4^{2-}]$$
$$K_{sp_2} = [Ba^{2+}][CO_3^{2-}]$$
$$\frac{K_{sp_1}}{K_{sp_2}} = \frac{SO_4^{2-}}{CO_3^{2-}} = \frac{1\times10^{-10}}{5\times10^{-9}} = 0.02$$

Q.30 Answer:

decreases, increases

Solution:

$$\text{Bond order} = \frac{10-4}{2} = 3$$
$$\text{MOEC of } N_2^+$$
$$= \sigma 1s^2, \sigma^{\star} 1s^2, \sigma 2s^2, \sigma^{*} 2s^2, \pi 2p_x^2$$
$$\equiv \pi 2p_y^2, \sigma 2p_z^1$$

Bond order $= \frac{9-4}{2} = 2.5$

Here, on converting N_2 to N_2^+ bond order decreases, so bond strength increases.

MOEC of $O_2 = \sigma 1s^2, \sigma^\star 1s^2, \sigma 2s^2, \sigma^\star 2s^2$

$\sigma p_z^2, \pi 2p_x^2 \equiv \pi 2p_y^2, \pi^* 2p_x^1 = \pi^* 2p_y^1$

Bond order $= \frac{10-6}{2} = 2$

MOEC of

$O_2^+ = \sigma 1s^2 \sigma^\star 1s^2, \sigma 2s^2 \sigma^* 2s^2, \sigma 2p_z^2$

$\pi 2p_x^2 \equiv \pi 2p_y^2, \pi^* 2p_x^1 \pi^* 2p_x^1$

Bond order $= \frac{10-5}{2} = 2.5$

On conversion of O_2 to O_2^+ bond order increases and hence bond strength also increases.

Q.31 Answer:

A, B, C

Solution:

Given $\rightarrow E^0_{A^3+/A} = 1.50\ \text{V}; E_{B^{2+}/B} = 0.3\ \text{V}; E^0\text{C}^{3\pi/\text{h}} = 0.74\ \text{V}\ E_{D^2/D} = -2.37\ \text{V} \quad \rightarrow$ these all are reduction potentials

We know that, reductien occurs at cathode in an electrolysis.

$\Rightarrow$ The more is the reduction potential, the more is the depositien of metal at cathode.

But, Cation having E° (Reduction potential) value less than $-0.83\ \text{V}$ (Reductien potential of H_2O) will not deposit from aqueous solution.

Here, $E^\circ_{A^{3+}}/A > E^\circ_{C^{3+}/c} > E^\circ_{b^{2+}/b}$

So, the rate of deposition world be ACB

$E_{D^{2+}/D} < -0.83\ \text{V}$ It will not deposit

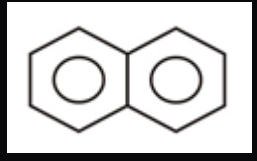
Q.32 Answer:

Assertion is True, Reason is False.

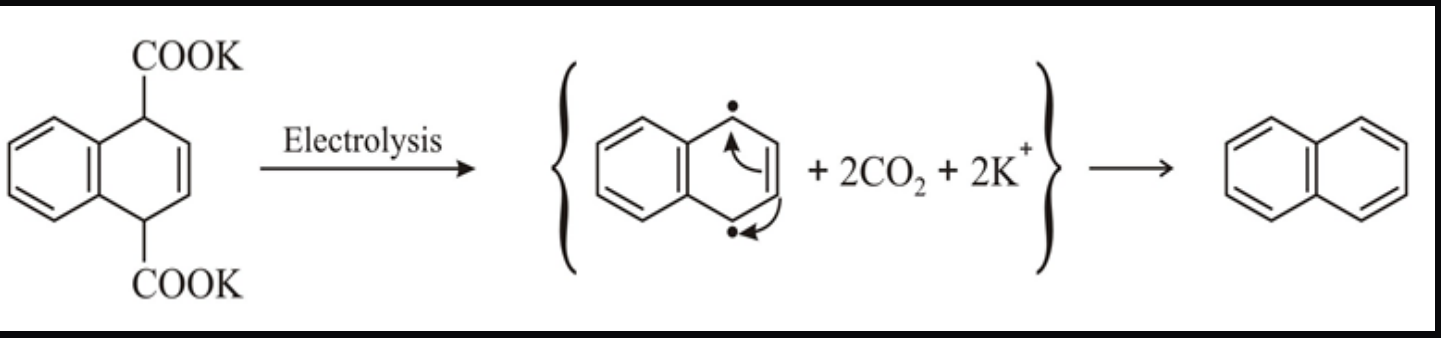
Solution:

CH_3CO- group in acetophenone being electron withdrawing reduces the electron-density at the benzene ring, thereby preventing further electrophilic substitution.

Q.33 Answer:



Solution:



Q.34 Answer:

Both assertion and reason statements are true but the reason is not the correct explanation of the assertion.

Solution:

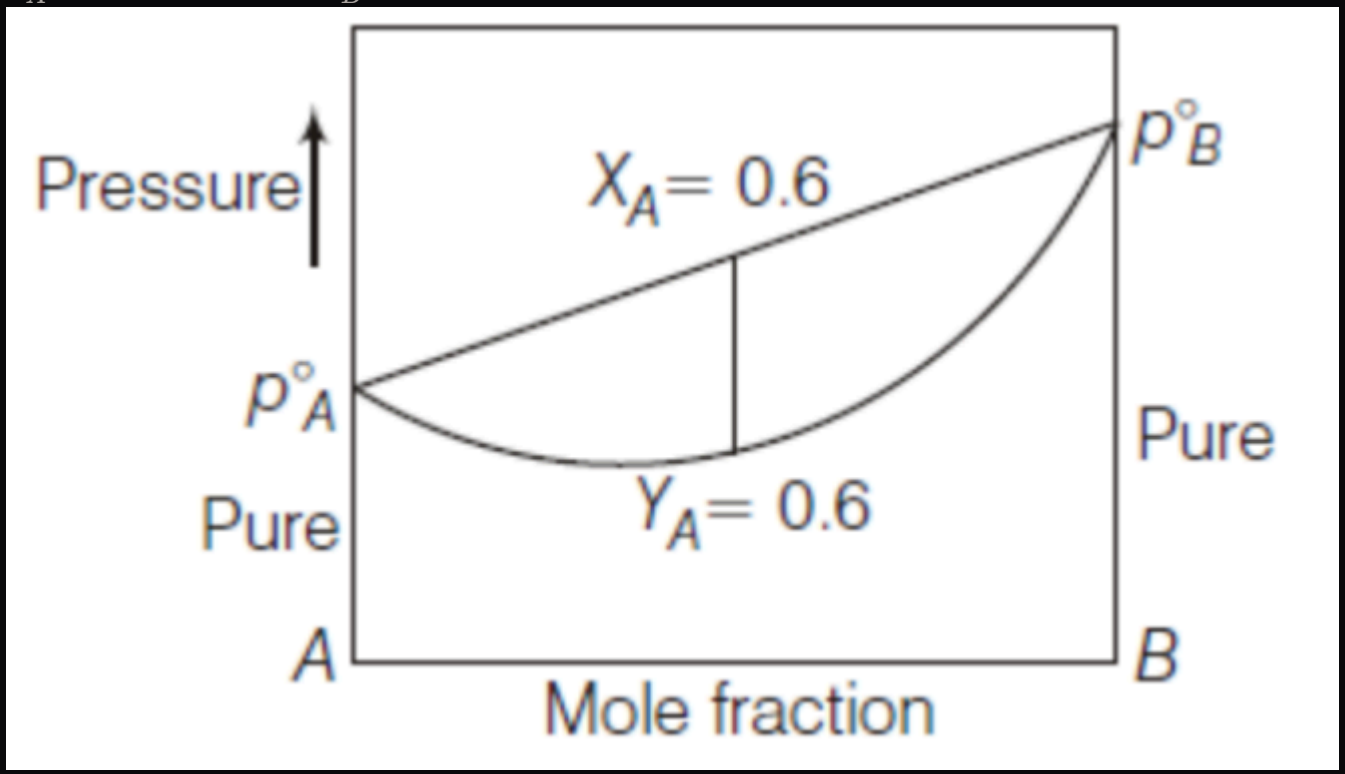
All lanthanoids belonging to the 3rd group of the periodic table do not justify why Lanthanum is called a d-block metal. It is called a d-block metal because the differentiating electron enters the d-orbit.
Hence, both the assertion and reason statements are true but the reason is not the correct explanation of the assertion.

Q.35 Answer:

$X_A = 0.74, X_B = 0.256$

Solution:

$p_A^\circ = 1000\ torr$ **and** $p_B^\circ = 1600\ torr$



$X_A = 0.6 \quad Y_A = 0.6$ **(from graph)**

$Y_A = \frac{p_A^\circ X_A}{p_A^\circ X_A + p_B^\circ (1 - X_A)}$

$0.6 = \frac{1000 X_A}{1000 X_A + 1600(1 - X_A)}$

$X_A = 0.70 \quad X_B = 1 - X_A = 0.30$

When last drop of liquid is remaining then, vapour has almost same composition as liquid in the start.

i.e. $y_A = 0.6$

$y_B = 0.4$

$\therefore \quad P_T.y_A = P_A^0 x_A$

$1240 \times 0.6 = 1000 \times x_A \Rightarrow x_A = 0.744$

$\therefore x_B = 0.256$

Q.36 Answer:

p q r s

3 1 5 2

Solution:

Compounds (I) and (II) both contain complex cation and anion, so they show coordination isomerism.

Compounds (III) and (IV) are obtained by the interchange of groups outside the coordination sphere and the ligands within the coordination sphere. Therefore, they given different ions in the solution. Hence, they show ionization isomerism.

Compounds (V) and (VI) are linkage isomers, because the ambident ligands, SCN has two donor atoms, i.e., it has two possibilities of attachment (SCN and NCS) with the central metal. Hence, they show linkage isomerism.

Compounds (VII) and (VIII) are obtained by the replacement of a coordinated group such as Cl^- by water of hydration. Hence, they show hydrate isomerism.

Q.37 Answer:

S^{2-}

Solution:

Ionization Energy:

Energy required to remove an electron from atoms or ions is called ionization energy.

Ionization energy is inversely proportional to size.

S^{2-} has the largest size and hence, has the lowest ionisation energy

Q.38 Answer:

17.46

Solution:

Determination of volume of nitrogen

Given, mass of substance $= 0.30$ g

V_{N_2} (volume of N_2) $= 50\ cm^3$

Atmospheric pressure $= 715$ mm Hg

Room temperature $= 300\ \mathrm{K}$ Vapour pressure of water at $300\ \mathrm{K}$

$= 15\ \mathrm{mm\ Hg}$

Actual vapour pressure of dry gas

$= 715 - 15 = 700\ \mathrm{mm\ Hg}$

Now to determine the volume of $\mathrm{N_2}$, we use

$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2}$

$\frac{700\times50}{300} = \frac{760\times V_2}{273}$

$V_2 = \frac{273\times700\times50}{300\times760} = 41.9\ \mathrm{cm^3}$

The mass of nitrogen in $41.9\ \mathrm{cm^3}$ gas is calculated by using concept of gram molar volume.

$22400\ \mathrm{cm^3}$ of $\mathrm{N_2}$ at STP weighs $= 28\ \mathrm{g}$

$41.9\ \mathrm{cm^3}$ of $\mathrm{N_2}$ at STP weigh

$= \frac{28\times41.9}{22400}\ \mathrm{g}$

Percentage of $\mathrm{N_2} = \frac{\text{Mass of } \mathrm{N_2}}{\text{Total mass of organic compound}} \times 100$

$= \frac{28\times41.9}{22400} \times \frac{100}{0.3} = 17.46\%$

Q.39 Answer:

2−Iodobutane, Butan−2−ol, 2 isomers

Solution:

When 2−bromobutane is treated with NaI in the presence of dry acetone then, iodine is substituted in the place of the bromine atom and formation of 2−iodobutane (B) takes place. This reaction is known as the Finkelstein reaction.

$$\underbrace{CH_3-CH_2-\underset{\underset{Br}{|}}{CH}-CH_3}_{(A)} \xrightarrow[\text{dry acetone}]{NaI} \underbrace{CH_3-CH_2-\underset{\underset{I}{|}}{CH}-CH_3}_{(B)}$$

2−Iodobutane reacts with moist $\mathrm{Ag_2O}$ to oxidise into corresponding alcohol, i.e., butan−2−ol (C).

$$\underbrace{CH_3-CH_2-\underset{\underset{I}{|}}{CH}-CH_3}_{(B)} \xrightarrow{Ag_2O} \underbrace{CH_3-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3}_{(C)}$$

Butan−2−ol is unsymmetrical and has one chiral carbon, therefore, the number of optical isomers $= 2^n$, where n =the number of chiral centres.

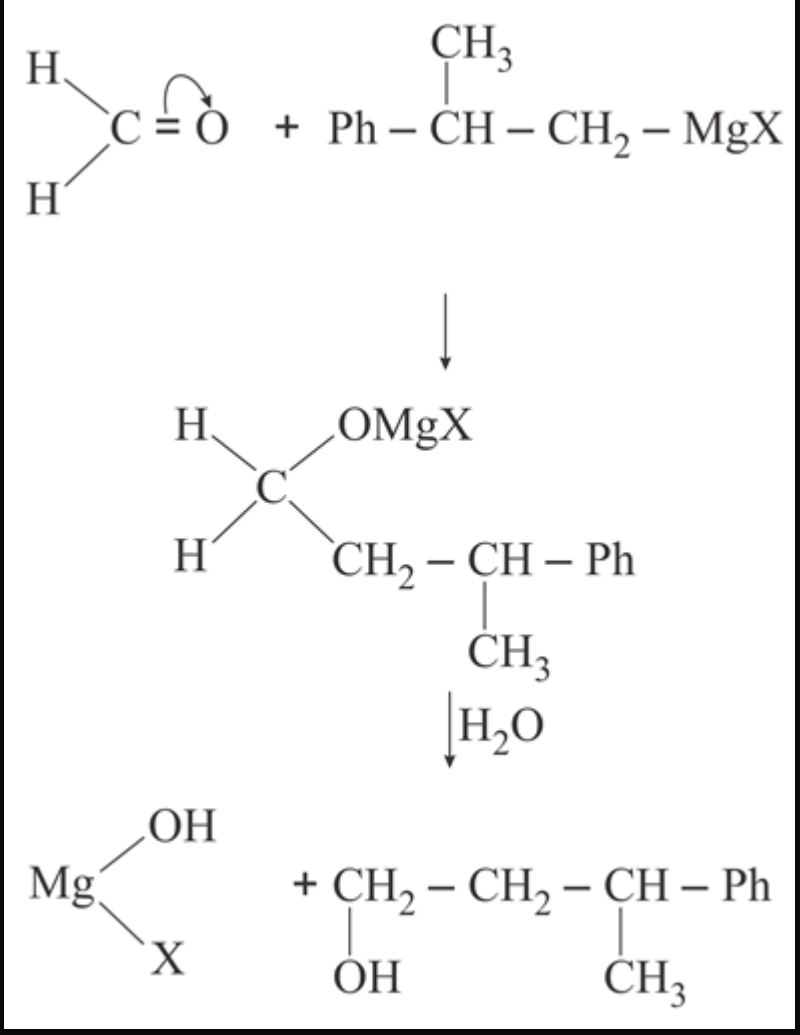
So, the number of optical isomers $= 2^1 = 2$.

Q.40 Answer:

$\mathrm{HCHO + PhCH(CH_3)\,CH_2\,MgX}$

Solution:

The given compound is tertiary alcohol and formaldehyde forms secondary alcohol on reaction with Grignard reagent.



Q.41 Answer:

$\Delta G < 0, \Delta S > 0, \Delta U = 0$

Solution:

$E_{a(f)} = E_{a(b)}$

$E_{a(f)} - E_{a(b)} = 0 = \Delta H(\Delta U)$

$\Delta G = \Delta H - T\Delta S$

$\Delta G = -T\Delta S$

For spontaneity ΔG is - ve

$\therefore$ ΔS is + ve.

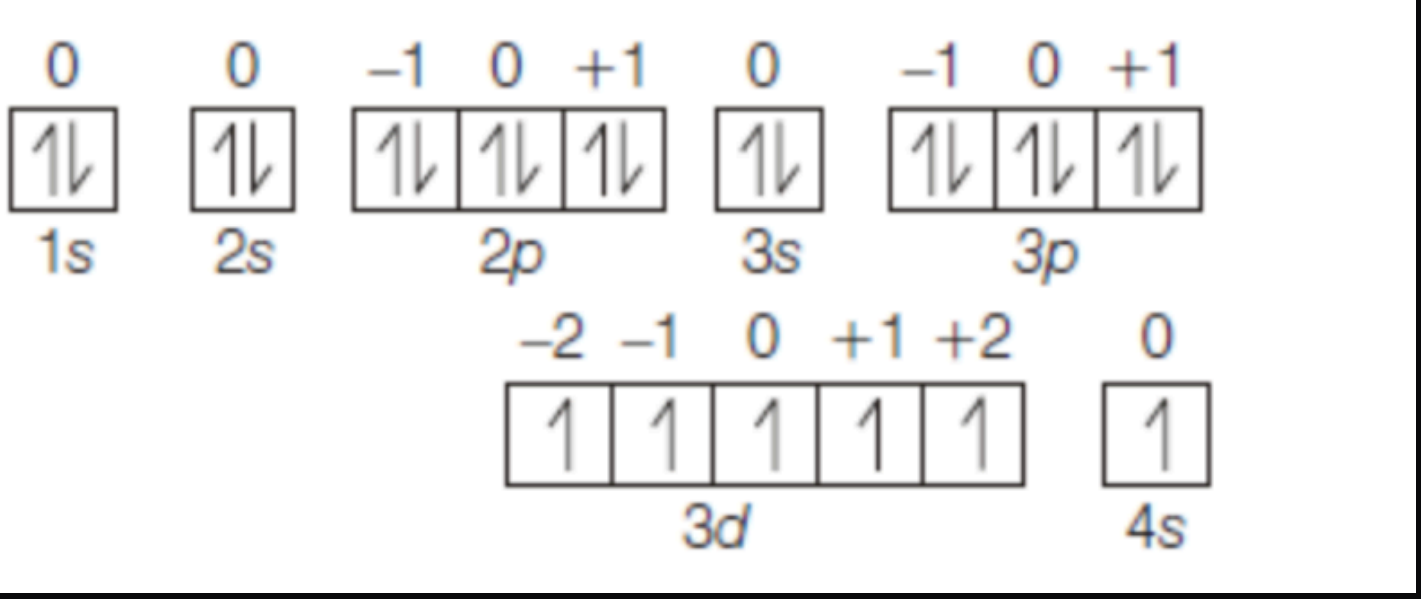
Q.42 Answer:

$2, 3$

Solution:

Atomic number (Z) of $Cr = 24$

Electronic configuration $= 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$



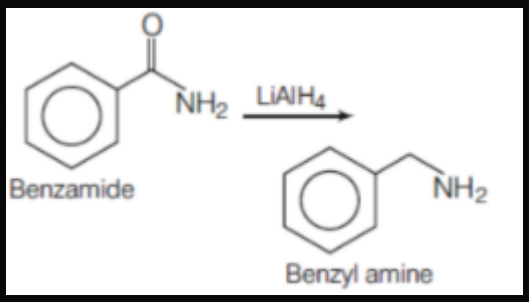
Among 6 electrons in $2p$ and $3p$, one electron must have $m = +1$ and $s = -\frac{1}{2}$ but in $3d$-subshell, an orbital having $m = +1$ may have spin quantum number $-\frac{1}{2}$ or $+\frac{1}{2}$.
Therefore, minimum and maximum possible values of electrons are 2 and 3 , respectively.

Q.43 Answer:

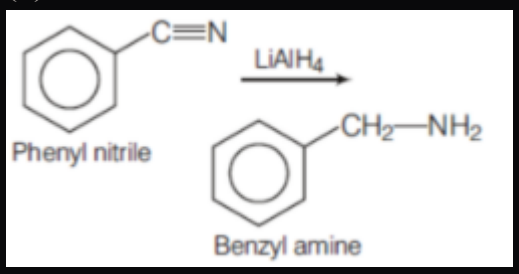
2-phenylethyl isonitrile $\xrightarrow{LiAlH_4}$

Solution:

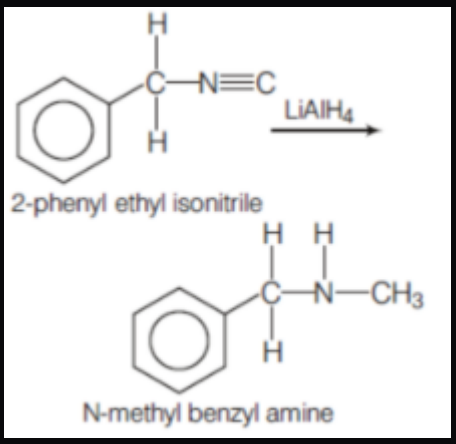
(a)



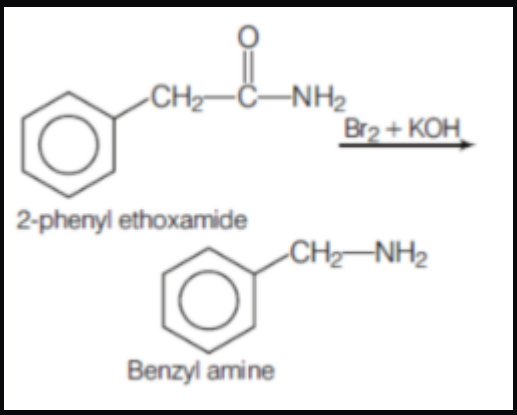
(b)



(c)



(d)



Last process is a good example of Hoffinann bromamide reaction in which carboxylamide are converted into amine having one carbon atom less than carboxyl amide.

Q.44 Answer:

All of the above

Solution:

$\mathrm{Al} + 6\ \mathrm{F}^{-} \longrightarrow \mathrm{AlF}_6^{3-}$

$H = \frac{3+6+3}{2} = 6$

Hybridisation $= sp^3d^2$ **structure**

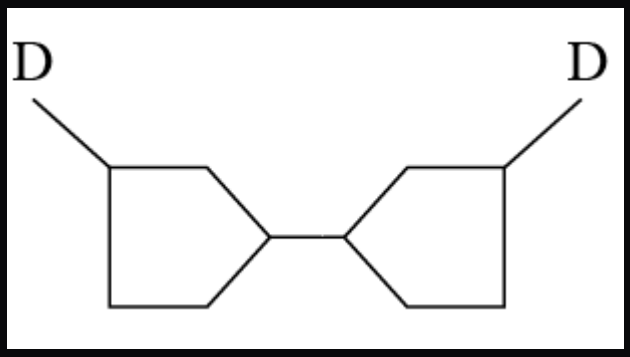
$=$ **octahedral** $\left[\mathrm{Be}\left(\mathrm{C}_2\mathrm{O}_4\right)_2\right]^2$

$H = \frac{2+4+2}{2} = 4$

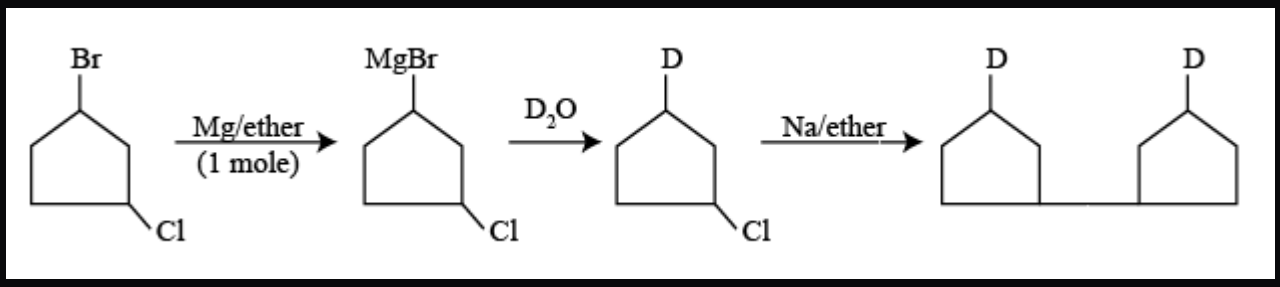
Hybridisation $= \mathrm{sp}^3$

Structure = tetrahedral

Q.45 Answer:



Solution:



Q.46 Answer:

—

Solution:

$$\mathrm{pOH}=\mathrm{pK_b}+\log\frac{[\text{salt}]}{[\text{base}]}$$
$$=-\log 10^{-10}+\log 1=10;\mathrm{pH}=14-10=4$$

Q.47 Answer:

—

Solution:

$$\mathrm{H_2(\,g)}+\tfrac{1}{2}\mathrm{O_2(\,g)}\rightarrow\mathrm{H_2O(g)}\quad\Delta\mathrm{H}=-249$$

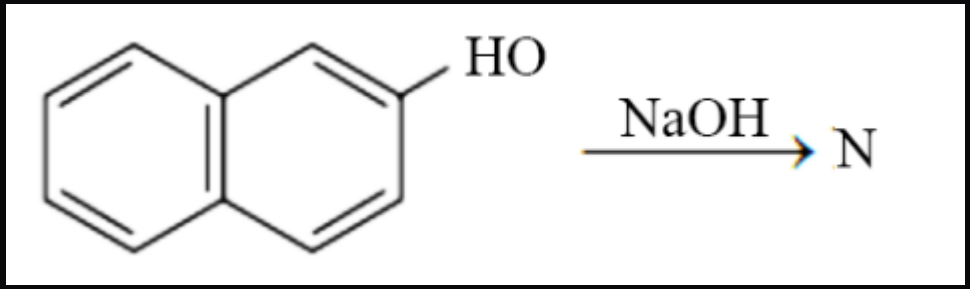
Let the bond enthalpy of $\mathrm{O-H}$ is x. Then $\Delta\mathrm{H}=\Sigma$ B.E. of reactant $-\Sigma$ B. E.of product

$$-249=433+\tfrac{1}{2}\times 492-2\mathrm{x}$$
$$\Rightarrow \mathrm{x}=464\ \mathrm{kJ\ mol^{-1}}$$

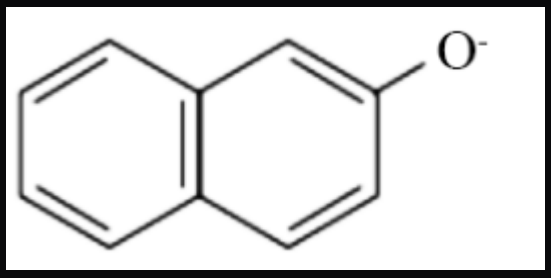
Q.48 Answer:

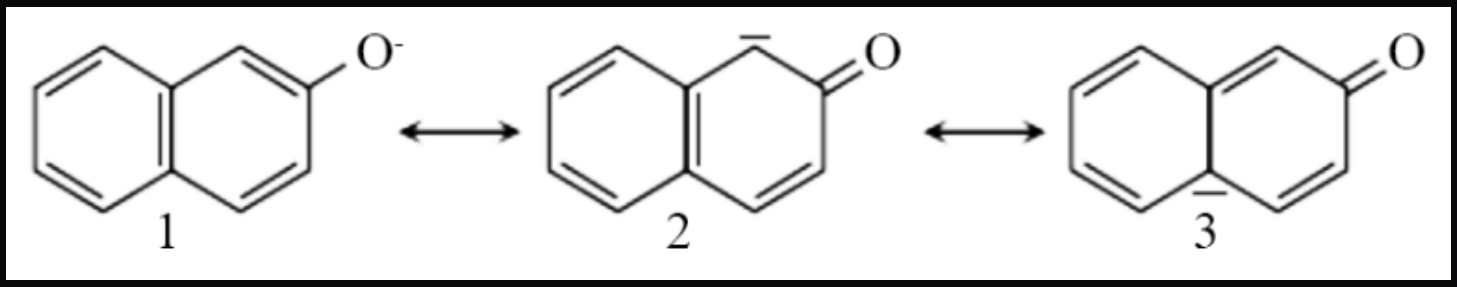
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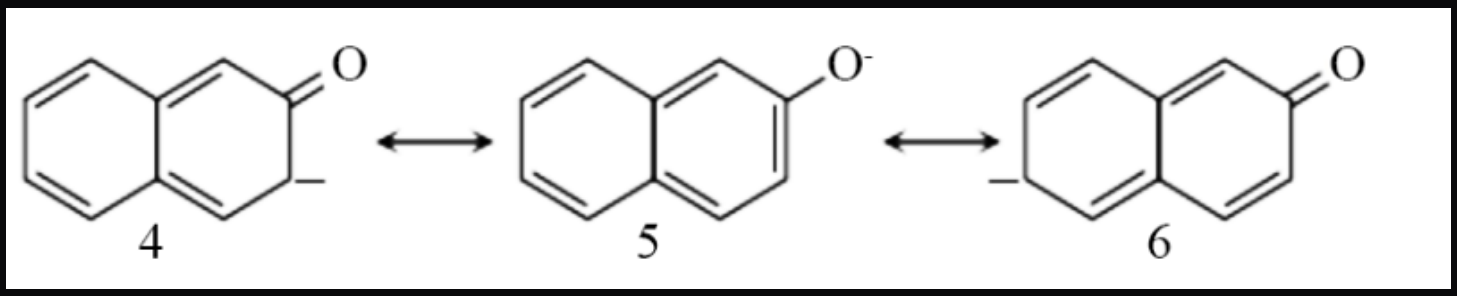
Solution:

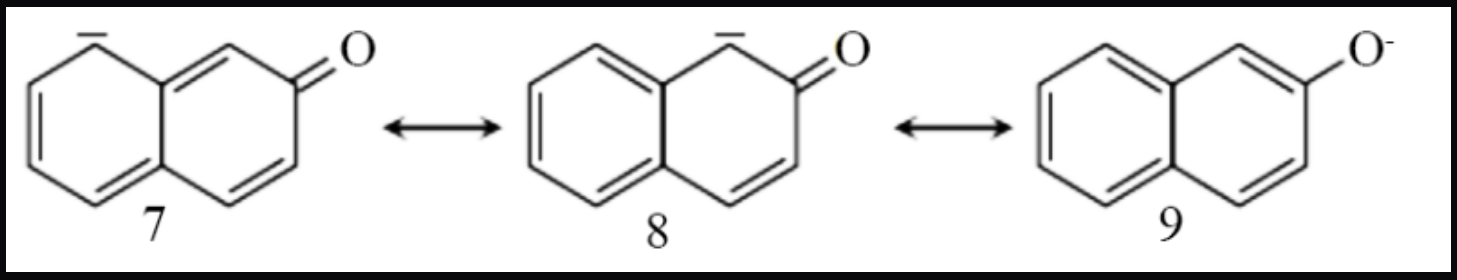


N is









Q.49 Answer:

—

Solution:

The balanced chemical equation for the oxidation of the complex is as given below.

$$MnO_4^- + 5\left[Fe\left(H_2O\right)\left(C_2O_4\right)_2\right]^{2-} + 8H^+ \rightarrow 5\left[Fe\left(H_2O\right)\left(C_2O_4\right)_2\right]^- + Mn^{2+} + 4H_2O$$

Hence, $-\frac{d}{dt}\left[MnO_4^-\right] = -\frac{1}{8}\frac{d}{dt}\left[H^+\right]$

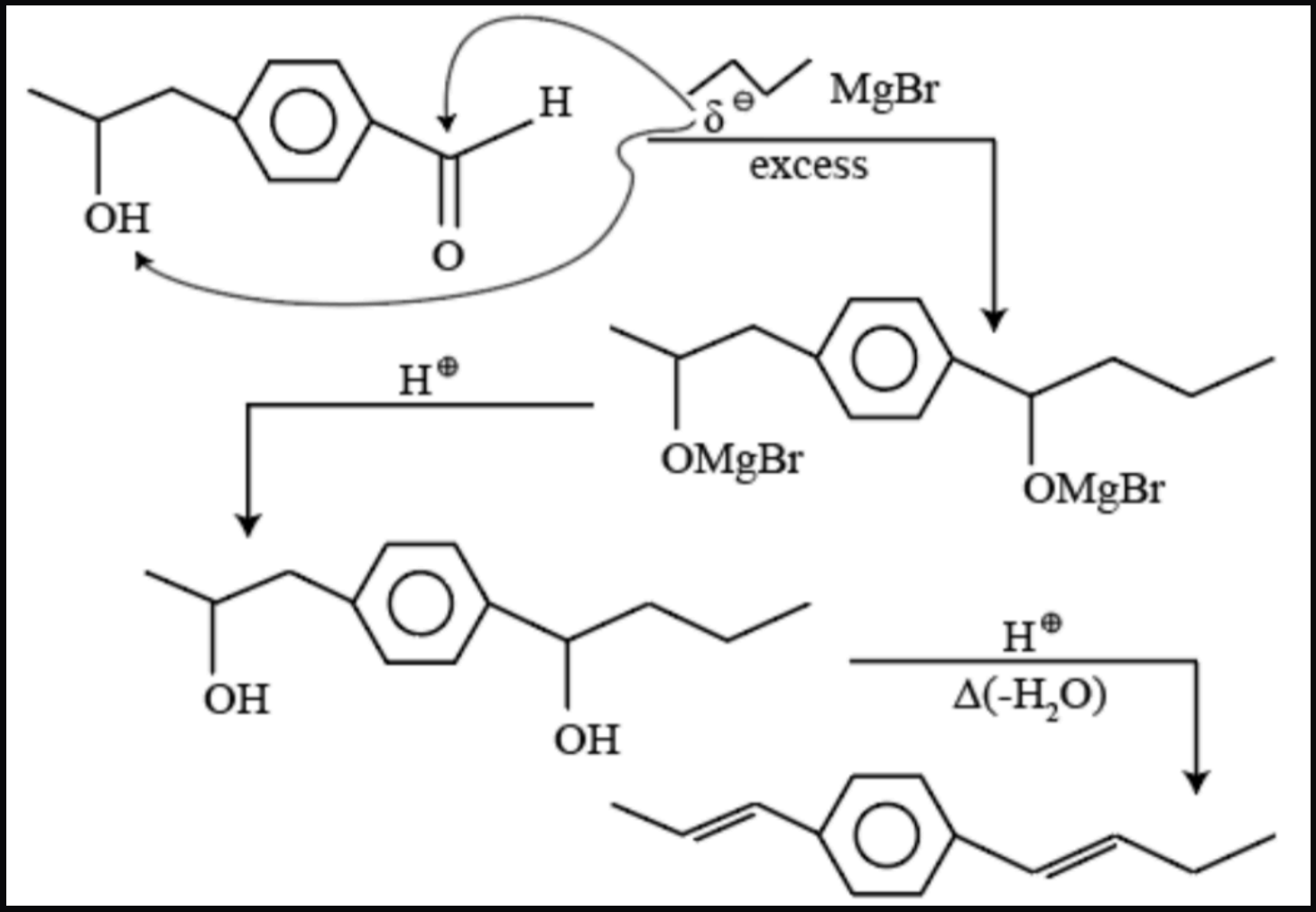
$$\frac{\frac{d}{dt}\left[H^+\right]}{\frac{d}{dt}\left[MnO_4^-\right]} = 8$$

Hence, the ratio of the rate of change of $\left[H^+\right]$ to the rate of change of $\left[MnO_4^-\right]$ is 8.

Q.50 Answer:

—

Solution:



Here the product 'A' has two double bonds with different groups so it has four geometrical isomers.

Q.51 Answer:

$\frac{5}{24}$

Solution:

A_1 : P **rides** A_1

A_2 : Q **rides** A_1

A_3 : R **rides** A

E : **Horse** A **wins**

$$\mathrm{P}\left(\mathrm{A}_1\right)=\frac{3}{6}, \mathrm{P}\left(\mathrm{A}_2\right)=\frac{2}{6}, \mathrm{P}\left(\mathrm{A}_3\right)=\frac{1}{6}$$

$$\mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{A}_1}\right)=\frac{1}{8}, \mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{A}_2}\right)=2 \times \frac{1}{8}, \mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{A}_3}\right)=3 \times \frac{1}{8}$$

$$\mathrm{P}(\mathrm{E})=\sum_{\mathrm{i}=1}^{3}\left[\mathrm{P}\left(\mathrm{A}_{\mathrm{i}}\right) : \mathrm{P}\left(\frac{\mathrm{E}}{\mathrm{A}_{\mathrm{i}}}\right)\right]$$

$$=\left(\frac{3}{6} \times \frac{1}{8}+\frac{2}{6} \times \frac{2}{8}+\frac{1}{6} \times \frac{3}{8}\right)=\frac{5}{24}$$

Q.52 Answer:

1

Solution:

$$\begin{aligned}&\because \quad f(0)=2 \Rightarrow f(f(0))=f(2)=8\\&\Rightarrow \quad f(f(f(0)))=f(8)=4\\&\Rightarrow \quad f(f(f(f(0))))=f(4)=0\end{aligned}$$

Q.53 Answer:

$\frac{8}{9}$

Solution:

$$\begin{aligned}&\left|\cdot A^{-1}\cdot \operatorname{adj}\left(B^{-1}\right)\cdot \operatorname{adj}\left(2A^{-1}\right)\right|=\frac{1}{|A|}\cdot\frac{1}{|B|^2}\cdot\left|2A^{-1}\right|^2\\&=\frac{1}{|A|}\cdot\frac{1}{|B|^2}\cdot\frac{64}{\left|A^2\right|^2}=\frac{64}{|A|^3|B|^2}=\frac{64}{8\times 9}=\frac{8}{9}\end{aligned}$$

Q.54 Answer:

1

Solution:

Misplaced \hline

$$\begin{aligned}&2x^2+(a+b)x+27=0\rightarrow(1)\\&x^2+(a+b)x+36=0\rightarrow(2)\\&(1)-(2)\Rightarrow x^2-9=0\Rightarrow x=\pm 3;+\text{ve root is } x=3\\&a=-7,b=-8\Rightarrow a-b=1\end{aligned}$$

Q.55 Answer:

2

Solution:

$$\begin{aligned}&\left(1-x^3\right)^5\left(1+x^2\right)^4\left(1+x^4\right)^8\\&\left[{}^5C_0-{}^5C_1x^3+{}^5C_2x^6\ldots..\right]\left[{}^4C_0+{}^4C_1x^2+{}^4C_2x^4+{}^4C_3x^6+{}^4C_4x^8\right]\\&\left[{}^8C_0+{}^8C_1x^4+\ldots.\right]\end{aligned}$$

Coefficient of x^6

$$\begin{aligned}&={}^5\mathrm{C}_0{}^4\mathrm{C}_3{}^8\mathrm{C}_0+{}^5\mathrm{C}_0{}^4\mathrm{C}_1{}^8\mathrm{C}_1+{}^5\mathrm{C}_2{}^4\mathrm{C}_0{}^8\mathrm{C}_0\\&=4+32+10=46\end{aligned}$$

Q.56 Answer:

2

Solution:

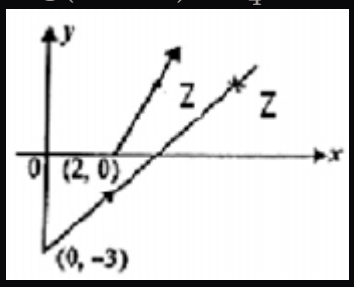
The line $\frac{x}{p}+\frac{y}{q}=1$ where $'p'$ is a prime
and $'q$ is a positive integer passes through $^{(4,3)}$
only if $q=\frac{3p}{p-4}=3-\frac{12}{p-4}$.
Since, $'$ ' is a positive integer, $p>4$ and $(p-4)$ divides 12,
there are only two such primes,
$p=5$ and $z=7$

Q.57 Answer:

0

Solution:

$\arg(z+3i)=\frac{\pi}{4}$ is a ray with slope 1 and starting from the point $(0,-3)$



$\arg(=-2)=\frac{\pi}{3}$ is a ray with slope $\sqrt{3}$ and starting from point $(2,0)$
The two rays do not intersect. There is no common z.

Q.58 Answer:

256

Solution:

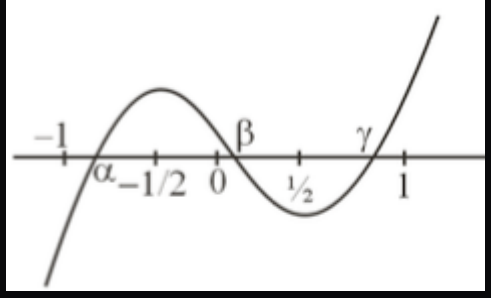
$770=2\times5\times7\times11$ We can assign 2 to x_1 or x_2 or x_3 or x_4
thus 2 can be assigned in 4 ways similarly each of $5,7,11$ can be assigned in 4 ways
$\therefore$ number of solutions $=4\times4\times4\times4=256$

Q.59 Answer:

$\sin^{-1}\gamma=\frac{7\pi}{18}$

Solution:

$$f(x)=8x^3-6x+1$$
$$f'(x)=6\left(4x^2-1\right)$$



Also $3x - 4x^3 = \frac{1}{2}$

put $x = \sin\theta$

$\sin 3\theta = \dfrac{1}{2}$

$\Rightarrow 3\theta = \dfrac{\pi}{6}, \dfrac{5\pi}{6}, \dfrac{13\pi}{6}, \dfrac{17\pi}{6}, \dfrac{25\pi}{6}, \dfrac{29\pi}{6}, \ldots$

$\Rightarrow \alpha = \sin\dfrac{25\pi}{18}, \beta = \sin\dfrac{\pi}{18}, \gamma = \sin\dfrac{5\pi}{18}$ or $\sin\dfrac{13\pi}{18}$

Q.60 Answer:

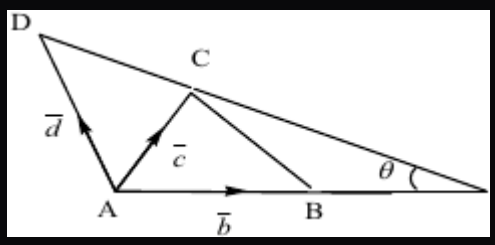
$\cos^{-1}\frac{2}{\sqrt{7}}$

Solution:

Let $AC = \lambda \Rightarrow AB = 5\lambda$ and $AD = 3\lambda$

$\overline{AB} = \bar{b}, \overline{AC} = \bar{c}, \overline{AD} = \bar{d}$

$\cos\theta = \dfrac{\overline{BA}\cdot\overline{CD}}{|\overline{BA}||\overline{CB}|}$



$= \dfrac{-\bar{b}(\bar{d}-\bar{c})}{|\bar{b}||\bar{c}-\bar{d}|}$

$= \dfrac{\bar{d}\cdot\bar{b} - \bar{b}\cdot\bar{c}}{5\lambda\cdot\sqrt{7}\lambda}$

$= \dfrac{10\lambda^2}{5\sqrt{7}\lambda^2}$

$\theta = \cos^{-1}\left(\dfrac{2}{\sqrt{7}}\right)$

Q.61 Answer:

$Z - \{0\}$

Solution:

$[x]$ is discontinuous for all integers.
The domain of $f(x) = \mathrm{R} - [-1, 0). f(x) = 0$ in $[0, 1)$
$\therefore f(x)$ is continuous at $x = 0$ and discontinuous for all integers $x \neq 0$ in the domain.

Q.62 Answer:

$\sqrt{2}$

Solution:

$I_1 = \int_0^{\frac{\pi}{4}} f(\cos 2x) \cos x dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} f(\cos 2x) \cos x dx$
Setting $x = \frac{\pi}{4} - t$ in the first integral and $x = \frac{\pi}{4} + t$ in the second integral,

$$I_1 = \int_0^{\pi/4} f(\sin 2t) \cos\left(\frac{\pi}{4} - t\right) dt + \int_0^{\pi/4} f(\sin 2t) \cos\left(\frac{\pi}{4} + t\right) dt$$

$$= \int_0^{\frac{\pi}{4}} f(\sin 2t) \left[\cos\left(\frac{\pi}{4} - t\right) + \cos\left(\frac{\pi}{4} + t\right)\right] dt$$

$$= \sqrt{2} \int_0^{\frac{\pi}{4}} f(\sin 2t) \cos t dt = \sqrt{2} I_2 \Rightarrow \frac{I_1}{I_2} = \sqrt{2}.$$

Q.63 Answer:

8

Solution:

Eccentricity of the hyperbola $x^2 - y^2 \sec^2\theta = 5$ is
$e_1 = \sqrt{\frac{1+\sec^2\theta}{\sec^2\theta}} = \sqrt{1 + \cos^2\theta}$
Eccentricity of the ellipse $x^2 \sec^2\theta + y^2 = 25$ is

$$e_2 = \sqrt{\frac{\sec^2\theta - 1}{\sec^2\theta}} = |\sin\theta| \text{ Given } e_1 = \sqrt{3} e_2$$

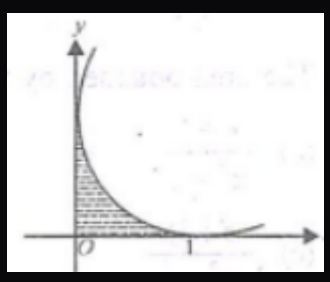
$$\Rightarrow 1 + \cos^2\theta = 3\sin^2\theta \Rightarrow \cos\theta = \pm\frac{1}{\sqrt{2}}$$

$\therefore$ Least positive value of θ is $\frac{\pi}{4}$ $\therefore P = 4 \Rightarrow 2P = 8$

Q.64 Answer:

$\frac{1}{6}$

Solution:



$$A = \int_0^1 y dx$$
$$= \int_0^1 (1 - \sqrt{x})^2 dx$$
$$= \int_0^1 (1 - 2\sqrt{x} + x) dx = 1 - \frac{4}{3} + \frac{1}{2} = \frac{1}{6}$$

Q.65 Answer:

45

Solution:

$$\sum_{i=1}^{10} \frac{(x_i - \bar{x})^2}{10} = \sigma^2 \ldots (1)$$
$$\frac{\sum_{i=1}^{9} (x_i + 10 - \bar{x})^2 + (x_{10} - 90 - \bar{x})^2}{10} = \sigma^2 \ldots (2)$$

From (1) and (2)

$$\frac{\sum_{i=1}^{10} (x_i - \bar{x})^2}{10} = \frac{\sum_{i=1}^{10} (x_i - \bar{x})^2 + 20 \sum_{i=1}^{9} (x_i - \bar{x}) - 180 (x_{10} - \bar{x}) + 100 \times 9 + 8100}{10}$$
$$2 \times \sum_{i=1}^{9} (x_i - \bar{x}) + 2 (x_{10} - \bar{x}) - 20 (x_{10} - \bar{x}) + 90 + 810 = 0$$
$$\Rightarrow 900 = 20 (x_{10} - \bar{x}) \Rightarrow x_{10} - \bar{x} = 45$$

Q.66 Answer:

$3\vec{a}$

Solution:

$$\vec{r} \times \vec{x} = \vec{y} \times \vec{x} \quad \Rightarrow \quad (\vec{r} - \vec{y}) \times \vec{x} = \overrightarrow{0} \quad \Rightarrow \quad \vec{r} = \vec{y} + \lambda \vec{x}$$
$$\vec{r} \times \vec{y} = \vec{x} \times \vec{y} \Rightarrow (\vec{r} - \vec{x}) \times \vec{y} = \overrightarrow{0} \Rightarrow \quad \vec{r} = \vec{x} + \mu \vec{y}$$
$$\vec{y} + \lambda \vec{x} = \vec{x} + \mu \vec{y}$$
$$(2\vec{a} - \vec{b}) + \lambda(\vec{a} + \vec{b}) = (\vec{a} + \vec{b}) + \mu(2\vec{a} - \vec{b})$$
$$\Rightarrow \quad 2 + \lambda = 1 + 2\mu, -1 + \lambda = 1 - \mu \quad \Rightarrow \quad \mu = 1, \lambda = 1$$

The point of intersection is $3\vec{a}$

Q.67 Answer:

$e^e + 1$

Solution:

By Leibnitz rule, $f'(x)=\frac{1-x}{x}-\ln\frac{x}{e}+f(x)=\frac{1}{x}-\ln x+f(x)$

$f'(x)=f(x)=\frac{1}{x}-\ln x$

It is linear d.e with I.F. $=e^{-x}$

Solution,

$$f(x)e^{-x}=\int\left(\frac{1}{x}-\ln x\right)e^{-x}dx=\int\frac{d}{dx}\left(\ln x\cdot e^{-x}\right)dx$$

$$=\ln x\cdot e^{-x}+c$$

$\therefore f(x)=\ln x+ce^{t}$

But, $f(1)=e\therefore c=1$

$f(x)=\ln x+e^{x}$

$f(e)=1+e^{e}.$

Q.68 Answer:

1

Solution:

$$\left(\text{cosec}^{-1}x-2\right)\left(\text{cosec}^{-1}x-\pi/6\right)\geq 0$$

$$\Rightarrow \text{cosec}-1x\leq\frac{\pi}{6}$$

$$\Rightarrow x\in(-\propto,-1]\cup[2,\propto)$$

Q.69 Answer:

18

Solution:

Let the A.P. be $a-3d,a-d,a+d,a+3d.$

The sum of the terms $=48\Rightarrow a=12$

$\frac{(12-3d)(12+3d)}{(12-d)(12+d)}=\frac{27}{35}\Rightarrow 35\left(16-d^{2}\right)=3\left(144-d^{2}\right)\Rightarrow d^{2}=4\Rightarrow d=\pm 2$

The numbers are $12-6,12-2,12+2,12+6$ **or** $12+6,12+2,12-2,12-6=$ $6,10,14,18$ **or** $18,14,10,6,\Rightarrow$ **largest term is 18.**

Q.70 Answer:

neither one-one nor onto

Solution:

We have $(\mathrm{f}-\mathrm{x})(\mathrm{x})=(\mathrm{f}(\mathrm{x})-\mathrm{g}(\mathrm{X}))=$

$$\left(f-g(x)=(f(x)-g(x))=\begin{cases}2x+3, & x\in \text{ rational}\\ 4x, & x \text{ eimational}\end{cases}\right.$$

As, $f\left(\frac{-3}{2}\right)=0=f\left(\frac{\sqrt{5}}{3}\right)$ **and so on.**

$\Rightarrow f(x)$ **is many one function.**

Also, $-\sqrt{5}$ **does not belong to the range, because if** $3x-\sqrt{5}=-\sqrt{5}$

$\therefore x = 0 \notin Q$

$\Rightarrow f(x)$ **Is into function**

Q.71 Answer:

—

Solution:

As $(3053)^{456} - (2417)^{333} = (339 \times 9 + 2)^{456} - (269 \times 9 - 4)^{333}$

Remainder of given number is same as remainder of $2^{456} + 4^{333}$ **and** $2^{456} + 4^{333} = (64)^{76} + (64)^{111}$

$= (1 + 63)^{76} + (1 + 63)^{111} = (1 + 9 \times 7)^{76} + (1 + 9 \times 7)^{111}$

Hence the remainder is 2.

Q.72 Answer:

—

Solution:

Required number of ways

= Total when all A's separated − Total when A's separated and H's are together

$$= \frac{7!}{2!}\left({}^8C_4\right) - 6!\left({}^7C_4\right)$$

$$= \frac{7! \cdot 6!}{4! \cdot 3!}(6) = 4^1 \cdot 5^2 \cdot 6^3 \cdot 7^1$$

Q.73 Answer:

—

Solution:

$$(-2, -6, 34);\ (-6, 7, 7)\ d = \left|\frac{\left(\vec{a}_2 - \overrightarrow{a_1}\right) \cdot \vec{n}}{|\vec{n}|}\right| \quad ; \quad \vec{n} = \overrightarrow{n_1} \times \vec{n}_2$$

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -10 \\ 4 & -3 & -2 \end{vmatrix} = -36\hat{i} - 36\hat{j} - 18\hat{k} = -18\underbrace{(2\hat{i} + 2\hat{j} + \hat{k})}_{\vec{n}}$$

$$d = \left|\frac{(-4\hat{i} + 13\hat{j} - 27\hat{k}) \cdot (2\hat{i} + 2\hat{j} + \hat{k})}{\sqrt{2^2 + 2^2 + 1^2}}\right|$$

$$= \left|\frac{-8 + 26 - 27}{3}\right| = \left|\frac{-9}{3}\right| = 3$$

Q.74 Answer:

—

Solution:

$$\text{AP} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p^{-} \\ q_{-} \end{bmatrix} = \begin{bmatrix} ap+bq \\ cp+dq \end{bmatrix}$$

Given $AP = P$

$$\begin{bmatrix} ap+bq \\ cp-dq \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$$

Hence $ap+bq=p$ **...(1)**

And $cp+dq=q$ **...(2)**

$$(a-1)p+bq=0$$

$$cp+(d-1)q=0$$

for non trivial solution

$$\begin{vmatrix} a-1 & b \\ c & d-1 \end{vmatrix}$$

$$(\text{a}-1)(\text{d}-1)-\text{bc}=0$$

$$(\text{ad}-\text{bc})=\text{a}+\text{d}-1=5050-1=5049$$

hence the answer is 9

Q.75 Answer:

—

Solution:

Replacing c_1 **by** $c_1+c_2+c_3$

$$\begin{vmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{vmatrix} = (1+2x)\begin{vmatrix} 1 & x & x \\ 1 & 1 & x \\ 1 & x & 1 \end{vmatrix} = (1+2x)\left[1-x^2-x(1-x)+x(x-1)\right]$$

$$= (1+2x)\left(1-2x+x^2\right) = 1-3x^2+2x^3$$

$$\begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix} = (3-x)\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix}$$

$$= (3-x)\left[x^2-2x+x+x\right] = (3-x)x^2 = 3x^2-x^3$$

$\therefore$ **The equation becomes** $1-3x^2+2x^3+3x^2-x^3=0 \Rightarrow x^3+1=0$

$$\therefore \quad x=-\omega,-\omega^2, x=-\omega \Rightarrow x^{2011}+\frac{1}{x^{2011}}$$

$$= -\left(\omega^{2011}+\frac{1}{\omega^{2011}}\right) = -\left(\omega+\frac{1}{\omega}\right) = -\left(\omega+\omega^2\right) = 1$$

$$x=-\omega^2 \Rightarrow x^{2011}+\frac{1}{x^{2011}} = -\left(\omega^{4022}+\frac{1}{\omega^{4022}}\right) = -\left(\omega^2+\frac{1}{\omega^2}\right) = -\left(\omega^2+\omega\right) = 1$$